

Thermal-property profiles from well-logs in sedimentary rocks: a Novel Machine-Learning-based prediction tool trained on physically modelled synthetic data

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ABSTRACT

Thermal properties such as thermal conductivity (TC), thermal diffusivity (TD), and specific heat capacity (SHC) are essential for understanding and modelling the subsurface thermal field. In sedimentary basins, these parameters play a key role in characterizing the present-day thermal state and predicting its evolution, for example, in response to future geo-energy utilizations. Given the wide range of potential geo-energy utilizations and the frequent lack of sufficient sample material, many studies have focused on developing accurate prediction approaches. Machine learning (ML) offers promising non-linear statistical methods to enhance the mapping of interrelations between standard geophysical well-log readings and thermal rock properties. In this study, we introduce an open-access tool for computing profiles of thermal rock properties from standard geophysical borehole logging data, building upon and extending previous petrophysical studies. The tool employs various machine-learning approaches trained on large, physically modelled synthetic datasets that account for mineralogical and porosity variability across major sedimentary rock groups (clastic rocks, carbonates, and evaporates). It establishes functional relationships between thermal properties and different combinations of standard well-log data, including sonic velocity, neutron porosity, bulk density, and the gamma-ray index. We trained four different models including linear regression, AdaBoost, Random Forest, and XGBoost using 80% of the synthetic group data for model development, including training and hyperparameter tuning through cross-validation. The remaining 20% was held out as an independent test set for statistical validation, feature recognition, and input variable importance analysis. A total of 15 input log combinations (including all one, two, three, and four well-log configurations) were evaluated across four machine learning models (linear regression, AdaBoost, Random Forest, and XGBoost), resulting in 180 trained models. The model's predictive accuracy and reliability were further evaluated against independent laboratory drill-core measurements reported in recent studies. Our results indicate that the best-performing predictive models vary depending on the available log-combinations. However, XGBoost frequently outperforms other models in sedimentary rocks. When at least two well logs are provided as input variables, the best-performing models predict thermal conductivity with an uncertainty below 10% relative to borehole validation data (with laboratory-measured thermal conductivity). In most tested model cases and for most input log combinations, predictive errors for thermal conductivity range between 10 and 30% at the (point measurement) sample scale (cm to half a meter). However, when averaged over geological formations or borehole intervals (tens to thousands of meters), the accuracy of thermal-conductivity predictions improves significantly, with uncertainties of the interval

mean conductivity dropping below 5% for large intervals. For specific heat capacity, prediction accuracy for the best-performing models at the measurement scale is typically better than 5%. Thermal diffusivity reflects a larger variation, accumulating the uncertainties from conductivity and heat capacity. The presented log-based Python prediction tool provides an automated means to compute thermal parameters using the most suitable ML model for given well-log inputs, facilitating enhanced thermal characterization in sedimentary settings. This has practical relevance for geothermal or hydrocarbon exploration, or subsurface storage projects.

Keywords: Downhole methods, Heat flow, Machine learning, Sedimentary basin processes, Heat generation and transport, Europe

1. Introduction

A detailed understanding of thermal properties – such as thermal conductivity (TC), thermal diffusivity (TD), and specific heat capacity (SHC) – is fundamental for modelling heat transfer processes in geosystems. These properties play a crucial role in various applications, including (i) resource exploration, (ii) geothermal modelling studies, (iii) geotherm computations and deep thermal regime assessment, and (iv) reconstructing the thermal history of sedimentary basins (e.g., Labus & Labus, 2018; Schön, 2011; Waite et al., 2007). However, direct measurements of thermal properties remain challenging and costly, particularly in non-coring drilling operations or when dealing with heterogeneous and anisotropic rock types. As a result, numerous studies have explored alternative methods for deriving thermal properties through petrophysical interrelations. Relations have been offered in particular for density and sonic velocity (e.g., Anand et al. 1973; Poulson et al. 1981; Kukkonen & Peltoniemi 1998; Popov et al. 2003; Hartmann et al. 2008; Gegenhuber & Schön 2012). For a detailed review and further references, we refer to Abdulagatova et al. (2009) and Fuchs and Förster (2014).

One widely adopted approach for estimating thermal properties is based on well-log data. Well-logging operations are standard in most drilling projects, providing valuable insights into subsurface geology and rock properties. Many previous studies have developed empirical, regression-based methods to correlate TC with petrophysical properties like density, sonic velocity, porosity or shale content using linear relationships (cf. Poulson et al., 1981; Demongodin et al., 1991; Hartmann et al., 2005; Gegenhuber and Schön, 2012; Gąsior and Przelaskowska, 2014). However, such empirical relationships are often constrained by two main limitations. First, they are typically derived from datasets that reflect specific regional geological and petrophysical conditions, limiting their broader applicability. Second, the relationships between thermal and petrophysical properties are often non-linear, making linear regression-based approaches insufficient for capturing complex interdependencies. To overcome these challenges, more advanced non-linear predictive models are required.

In addition to empirical methods, theoretical models – commonly referred to as mixing laws – are also used to estimate thermal properties from well-log data (e.g., Bayuk et al., 2011; Chekhonin et al., 2021). These models rely on computing rock volume fractions and applying the corresponding thermal parameters of individual minerals or rock components. While these approaches provide a more physics-based estimation framework, their accuracy depends on the representativeness of the assumed mineralogical composition and volumetric fractions.

For a comprehensive overview of well-log-based prediction techniques and their respective advantages and limitations, we refer to Fuchs and Förster (2014). They critically assessed the transferability and applicability of empirical relations across different geological settings and proposed an alternative approach to mitigate the biases associated with regionally constrained empirical equations. Their methodology involved generating synthetic datasets for sedimentary rocks, reflecting a broad range of mineral compositions and petrophysical properties. These datasets enabled the derivation of more generalized prediction equations for matrix TC based on various standard well log inputs.

Building upon this framework, Fuchs et al. (2015) extended this approach by incorporating porosity variations (0–30%) to compute not only TC but also TD and SHC. They demonstrated that their synthetic database yielded superior predictive accuracy and applicability compared to fixed empirical regression-based methods, with reported error margins of 2–8 % for SHC, 5–10 % for TC, and 8–15 % for TD, respectively. Validation studies comparing predicted thermal profiles with laboratory-measured core conductivities and in-situ temperature gradient data further confirmed the reliability of their approach across various sedimentary basins in Europe (Fuchs et al., 2020).

In recent years, machine learning (ML) methods have gained increasing traction in petrophysical studies, demonstrating their ability to effectively model statistical variance in target parameters based on diverse input datasets (Mousavi & Hosseini-Nasab, 2024a). Various ML algorithms have been employed for different geophysical and reservoir characterization tasks, yielding significant improvements in predictive accuracy compared to traditional empirical and regression-based methods. For instance, Al-Mudhafar (2019) applied Bayesian Model Averaging (BMA) and Least Absolute Shrinkage and Selection Operator (LASSO) regression to model core permeability using well log and core data, achieving highly accurate predictions. Similarly, Allawi et al. (2023) introduced a novel semi-analytical model to estimate isothermal rock compressibility in sandstone formations, which was successfully validated with data from southern Iraq. Li et al. (2023) explored the use of ML algorithms, specifically the LightGBM model, to predict pore pressure in reservoir blocks with anomalously high pressures. Their approach achieved over 90% prediction accuracy, offering valuable insights for optimizing drilling efficiency and mitigating operational risks.

Several ML algorithms have been successfully employed in petrophysical studies, often outperforming traditional simple or multiple linear regression models. Artificial intelligence-based approaches have been applied to estimate key rock properties such as porosity (Chen et al., 2021), permeability (Elkatatny et al., 2018; Mathew Nkurlu et al., 2020), water saturation (Hadavimoghaddam et al., 2021; Miah et al., 2020), and lithology (Mousavi & Hosseini-Nasab, 2024b) using well-log data. For geothermal and thermal property estimations, various ML techniques, such as Support Vector Regression (SVR), Random Forest (RF), and Decision Tree (DT), have been explored (cf. Baghbani et al., 2023; Jaafreh et al., 2021; Otchere et al., 2022). Additionally, deep-learning (DL) methods, including convolutional neural networks (CNN), and artificial neural networks (ANN), have also been employed for thermal property prediction (Goutorbe et al., 2006; Pang et al., 2023; Sharo et al., 2022). However, most existing empirical studies rely on laboratory-derived data (cf. Samaei et al., 2022; Shakirov et al., 2021) rather than wireline borehole measurements, which are often recorded during exploration campaigns. Consequently, models trained on such datasets may have limited applicability beyond the specific geological conditions of the sampled rock formations, reducing their transferability to different sedimentary settings.

This study aims to address these limitations by integrating the synthetic data approach (Fuchs et al., 2015; Fuchs & Förster, 2014) with modern ML techniques to assess their effectiveness in predicting TC, TD, and SHC. Building upon prior research and extensive experience in applying synthetic prediction equations (e.g., Fuchs, 2018; Fuchs et al., 2020), we refine the classification of major sedimentary rock groups and optimize the selection of input variables. This study extends previous work by enhancing the physical modeling of synthetic petrophysical data for major sedimentary rock groups and incorporating recent advancements in ML methodologies.

This research highlights several key aspects: (i) The importance of generating synthetic rock datasets that represent sedimentary environments, including mineralogical variability, such as the influence of clay minerals in evaporite rock formations on rock porosity and effective thermal properties; (ii) The consideration of a broad porosity range for marine conditions, including clastic rocks and carbonates, with values extending up to 50% in low-compacted shallow sediments. This includes porosity increments of 5% for the bulk composition variability; (iii) A systematic evaluation of ML models for different well-log input combinations, assessing uncertainty levels for each sedimentary rock group. Finally, this study introduces ThermoProfiler, an open-source tool designed for the automated prediction of thermal rock properties (TC, TD, and SHC) in sedimentary formations based on standard geophysical well-logs inputs.

2. Methods and data

The workflow of this study comprises three main stages, as illustrated in Fig. 1.

In the first stage (**part I**), synthetic data were generated for three distinct sedimentary rock groups – carbonates, clastic rocks, and evaporates – using a modified version of the method introduced by Fuchs and Förster (2014). These datasets were designed to capture the mineralogical and petrophysical variability of sedimentary rocks, ensuring a robust basis for machine learning (ML) model training.

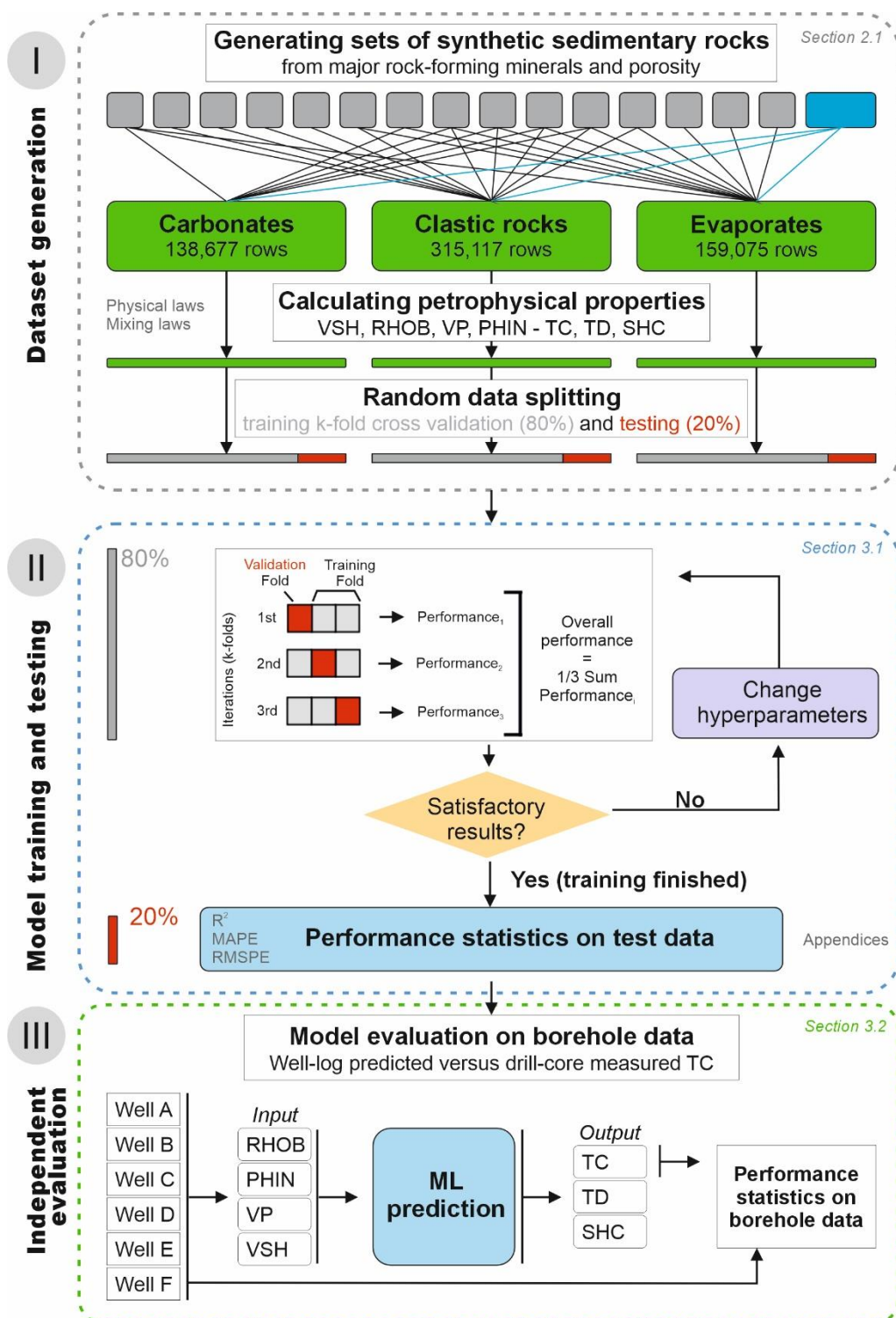


Fig. 1. Overview of the study workflow comprising three main stages: (I) generation of synthetic datasets for three sedimentary rock groups (clastic rocks, carbonates, and evaporates); (II) training and hyperparameter tuning of four machine learning models (linear regression, AdaBoost, Random Forest, and XGBoost) using k-fold cross-validation ($k = 3$) within the training dataset (80%), while the remaining 20% was held out as an independent test set; and (III) independent evaluation of the final trained models using measured borehole thermal conductivity (TC) data from sedimentary basins.

In the second stage (**part II**), four regression models – multi-linear regression, AdaBoost, RF, and XGBoost – were trained using 80% of the synthetic dataset as the training set. Each model was trained separately for the three main sedimentary rock groups, considering all possible well-log input combinations.

Hyperparameters were tuned using Bayesian optimization (Wu et al., 2019) for each log combination, employing a k-fold cross-validation method (k=3 considered) (Rao et al., 2008) to find the optimal set of hyperparameters and enhance the model reliability. This process involved splitting the dataset into equal subsets (folds) to independently train the model on each fold (Al-Mudhafar, 2016; Wang et al., 2014). Each model was trained three times, with two folds used for training and the remaining fold used for validation in each iteration. The final performance was averaged across the three runs to obtain a more reliable estimate of the model's generalization ability. Once the optimal hyperparameters were identified, the model was retrained on the entire training dataset (80% of the synthetic data) and evaluated on the remaining 20% of the synthetic data (test set) to assess their final predictive performance. Importantly, real-world data were not included in the training process; instead, they were exclusively for independent validation, ensuring that the models' generalization ability was assessed under realistic conditions. For each of the three rock groups, 15 predictor combinations were tested: 4 single-log, 6 two-log, 4 three-log, and 1 four-log combination. Each combination was modeled using four different machine learning techniques—linear regression, AdaBoost, Random Forest, and XGBoost—resulting in a total of 180 trained models.

The third stage (**part III**) involved independent validation using borehole TC data obtained from the literature. Since direct measurements for SHC and TD were unavailable, these parameters were indirectly evaluated based on inferred relationships and available data. The evaluation results on the synthetic dataset are reported in Section 3.1, while direct and indirect validation results based on real-world data are presented in Section 3.2.

2.1. Synthetic data

The synthetic datasets used in this study are generated based on the methodology introduced by Fuchs and Förster (2014), which reconstructs the composition of major rock-forming minerals within three primary sedimentary rock groups (clastic rocks, carbonates, and evaporates). For each mineral, individual minimum and maximum values are defined, allowing compositional variability within fixed ranges specific to each rock group (Table. 1). However, a reassessment of the original method revealed certain methodological limitations, which were addressed through the following modifications: (1) The original approach did not account for clay mineral and minor quartz content in evaporite formations (2) To better represent shallow marine sedimentary conditions, the porosity range for clastic rocks and carbonates was extended from 30% to 50%, with 5% incremental steps; new total range: 0–50%. (3) To prevent unrealistic mineral compositions, upper and lower bounds for each mineral class were refined. For synthetic carbonates and clastic rocks datasets, porosity values ranged from 0 and 50%, with a combination of porosity and mineral composition varying in 5% increments (Quartz100% - Porosity0%; Q95 - P05; Q90 - P10, etc.). In contrast, for evaporates, porosity ranged from 0 and 30%, with mineral combinations varying in 10% increments and porosity in 5% increments. These modifications resulted in a total of 138,677 synthetic samples for carbonates, 315,117 synthetic samples for clastic rocks, and 159,075 synthetic samples for evaporates, respectively.

Table 1. Compilation of minerals with respective varieties, employed in constructing the datasets for the main sedimentary rock groups.

Class	Mineral	Boundary	Range of mineral constitutes in		
			carbonates (%)	clastic rocks (%)	evaporates (%)
Oxides	Quartz	B1	0-30	0-100	0-10
	Anorthite	B2	-	0-50	-
Feldspars	Albite	B3	-	0-50	-
	Orthoclase	B4	-	0-50	-
Micas	Muscovite	B5	-	0-30	-
	Biotite	B6	-	0-30	-
Clays	Kaolinite	B7	0-70	-	0-40
	Montmorillonite	B8	0-70	0-100	0-40
	Illite	B9	0-70	0-100	0-40
Carbonates	Calcite	B10	0-100	0-20	0-100
	Dolomite	B11	0-100	0-20	0-100
Sulfates	Anhydrite	B12	-	0-20	0-100
	Gypsum	B13	-	-	0-100
Chlorides	Halite	B14	-	-	0-100
	Sylvite	B15	-	-	0-100

Once the synthetic datasets were generated, petrophysical properties, including standard well-log parameters (later used as input for predictions) and thermal properties (considered as target variables), were calculated. Petrophysical parameters later used as well-log input are bulk density (RHOB in g/cm³) from gamma-gamma density log, neutron porosity (PHIN in n/n) from neutron gamma log, the volume fraction of shale (VSH in n/n) interpreted as gamma-ray index from natural gamma logs GR and sonic velocity (VP in km/s) from acoustic log. The target variables (thermal properties for ML prediction) are TC in W/(m·K), TD in 10⁻⁶ m²/s, and SHC in J/(kg·K). To improve data consistency and model applicability, several refinements were made to the initial methodology: (1) The dataset was streamlined from five to four well log input variables, removing the photoelectric absorption index (U) for two reasons: it correlates with its bulk density input, and, it requires the rarely measured photoelectric factor (PEF) as well log input. (2) For consistency, we changed the unit for PHIN from percentage (%) to unitless (n/n) and used the sonic velocity (VP in km/s) instead of the sonic interval transit time (DT in μ s/m). (3) The mineral composition constraints for each rock group are modified. For carbonates, the cumulative content of calcite and dolomite (B10 and B11; cf. Table. 1) must be $\geq 50\%$, while clay minerals must remain $< 70\%$ (cf. B7, B8, and B9). For clastic rocks, the combined quartz and clay mineral content must be $\geq 50\%$ (cf. B1, B8, and B9), feldspars (B3 and B4) and micas (B5 and B6) must not exceed 40% each. Additionally, the total percentage of B10, B11, and B12 must remain $\leq 40\%$. For evaporites, the collective contribution of B7, B8, and B9 must not exceed 70% to ensure realistic compositions. For petrophysical computations, mineral end-member values as listed in Fuchs et al. (2015) were applied. The computed petrophysical dataset forms the basis for subsequent machine-learning predictions of TC, TD, and SHC.

The population statistics for the synthetic petrophysical parameter sets is listed in Table. 2. As expected, the three groups exhibit significantly different mean values and statistical distributions for most parameters, reinforcing the need for applying empirical and ML-based methods separately for each geologically distinct group.

Table 2. Population statistics for the synthetic training data sets (top: clastic rocks; middle: carbonates; bottom: evaporites).

Group	Variable	Mean	1-sigma std	Min	Median	Max
Clastic rocks	RHOB	2.24	0.25	1.61	2.24	2.87
	PHIN	0.32	0.16	-0.02	0.32	0.63
	VSH	0.55	0.21	0.00	0.60	1.00
	Vp	3.581	0.890	2.381	3.384	6.151
	SHC	1568	512	675	1568	2405
	TC	2.03	0.71	1.04	1.87	7.70
	TD	0.82	0.39	0.31	0.72	3.80
Carbonates	RHOB	2.28	0.25	1.76	2.28	2.88
	PHIN	0.32	0.16	-0.01	0.32	0.62
	VSH	0.29	0.12	0.00	0.30	0.50
	Vp	3.814	1.065	2.490	3.540	7.141
	SHC	1628	499	791	1626	2458
	TC	2.44	0.77	1.22	2.30	6.00
	TD	0.93	0.40	0.37	0.84	2.56
Evaporites	RHOB	2.29	0.18	1.76	2.28	2.93
	PHIN	0.25	0.11	-0.02	0.25	0.62
	VSH	0.28	0.17	0.00	0.30	0.70
	Vp	4.046	0.744	2.851	3.914	6.938
	SHC	1324	317	601	1327	1924
	TC	2.95	0.85	1.17	2.80	8.42
	TD	1.25	0.47	0.38	1.15	4.38

2.2. Machine learning models

In the presented study, we employed a multivariate linear regression model along with three ML models, which are AdaBoost, RF, and XGBoost. A brief description of these models is provided as follows.

2.2.1. AdaBoost

AdaBoost (Adaptive Boosting) is a powerful ensemble learning algorithm that has proven to be successful in enhancing the performance of weak learners, typically decision trees, to create a reliable and accurate predictive model (Freund & Schapire, 1997; Kummer & Najjaran, 2014). The basic concept behind AdaBoost is to allocate different weights to the observations in the training dataset, highlighting the incorrectly classified instances in each iteration. The weight of a weak learner (alpha) is calculated using the formula:

$$\alpha_t = \frac{1}{2} \ln \left(\frac{1 - \text{error}_t}{\text{error}_t} \right) \quad (1)$$

This adaptability allows AdaBoost to focus on the shortcomings of weak learners, sequentially combining their outputs to form a robust and accurate prediction. The algorithm assigns higher weights to misclassified instances, prompting the subsequent weak learners to prioritize them during their training (Géron, 2022). The final prediction

is then made by combining the weighted outputs of all weak learners as follows:

$$H(x) = \text{sign}(\sum_{t=1}^T \alpha_t h_t(x)) \quad (2)$$

where $H(x)$ is the final hypothesis, T is the number of weak learners, α_t is the weight of the t -th weak learner, and $h_t(x)$ is the prediction of the t -th weak learner.

2.2.2. Random Forest

Random Forest, proposed by Breiman (2001), is a powerful ML strategy that employs the principles of ensemble learning with decision tree models to make accurate predictions (Geurts et al., 2006). Unlike typical regression approaches that rely on a single model, random forest regression builds a multitude of decision trees and aggregates their outputs to improve overall prediction accuracy. The algorithm works by creating a random subset of the training data for each tree and choosing a random subset of features at each split. This randomness helps mitigate overfitting and increases the model's ability to generalize well to unseen data (Belyadi & Haghighat, 2021). The final prediction is then determined by averaging the predictions of all the distinct trees in the forest. The formula for predicting the target variable (Y) using random forest regression can be defined as follows:

$$\hat{Y} = \frac{1}{N} \sum_{i=1}^N h_i(x) \quad (3)$$

where $\hat{Y}$ is predicted target variable, N is the number of trees in the random forest and $h_i(x)$ is the prediction of the i -th tree in the forest for the input features x .

2.2.3. XGBoost

XGBoost (eXtreme Gradient Boosting) is a popular algorithm used for regression and classification problems (Liu et al., 2023; Zhang et al., 2020; Zhong et al., 2020). It is an ensemble algorithm that combines multiple weak models (decision trees) into a single robust model. The algorithm is based on the gradient boosting framework and applies the principle of boosting to enhance the accuracy of the model (Ayyadevara, 2018). The XGBoost regression model can be represented using the following equation:

$$y_i = \sum_{j=1}^M f_j(x_i) \quad (4)$$

The y_i is predicted target value of i th data point, and the model includes M trees. For the i th data point, $f_j(x_i)$ represents the predicted target value of the j th tree, and x_i represents the feature vector. XGBoost employs an optimization technique called gradient descent to reduce the loss function and find the optimal weights for the decision trees in the model. The loss function is a measure of how well the model can predict the target variable. The objective function of XGBoost can be represented as follows:

$$\text{obj} = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{j=1}^M \Omega(f_j) \quad (5)$$

where $l(y_i, \hat{y}_i)$ is the loss function for the i th data point, $\hat{y}_i$ is the predicted target value for the i th data point, and $\Omega(f_j)$ is the regularization term for the j th tree.

The regularization term is used to avoid overfitting and increase the model's generalization. The two widely used regularization terms in XGBoost are the L1 and L2 regularization terms. The L1 regularization term penalizes the absolute values of the weights, while the L2 regularization term penalizes the squared values of the weights (Chen & Guestrin, 2016). Overall, XGBoost is a robust algorithm for regression problems that can achieve high accuracy and handle large datasets with complex structures. Its potential to combine multiple weak models into a powerful model and its use of gradient descent for optimization make it a good choice for data modelling.

2.3. Statistical metrics and prediction accuracy assessment

The performance of the models was evaluated based on the following metrics: R-squared, mean absolute percentage error (MAPE), and root mean squared percentage error (RMSPE). R-squared, often denoted as R^2 is a statistical measure that represents the proportion of the variance in the dependent variable that is predictable from the independent variables. It is a valuable metric for assessing the goodness of fit of a regression model. For the ideal model, the value of R^2 is equal to one, and for MAPE and RMSPE, it is equal to zero.

$$R^2 = 1 - \frac{SSR}{SST} \quad (6)$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{M_i - \hat{M}_i}{M_i} \right| \times 100 \quad (7)$$

$$RMSPE = \sqrt{\frac{1}{n} \sum_{i=1}^n \left(\frac{M_i - \hat{M}_i}{M_i} \right)^2} \times 100 \quad (8)$$

where SSR is the sum of squared residuals, SST is the total sum of squares, representing the total variance in the dependent variable; and n is the number of observations, M_i is the observed value and $\hat{M}_i$ is the predicted value.

The importance of features in the applied machine learning models is calculated using the shapley additive explanations (SHAP) method. It was developed by Lundberg and Lee (2017) and has since become one of the most widely used methods for feature importance calculations. The main advantage of SHAP is that it provides a unified framework for interpreting different machine learning models. One of the critical benefits of the SHAP feature importance is its ability to handle complex machine-learning models. For example, traditional feature importance methods like permutation importance may struggle with models that have interactions between variables. SHAP, on the other hand, can handle these interactions and provide a more accurate estimate of feature importance. This makes it particularly useful for models like tree-based models and neural networks. Another benefit of SHAP is that it provides local and global feature importance measures. Local feature importance measures the impact of a specific feature on a particular prediction, while global feature importance measures the overall importance of each feature across all predictions. This allows us to identify which features are consistently important across the entire dataset and which features are only necessary in specific cases.

The SHAP values of a feature are the average marginal contributions of that feature across all possible coalitions of

features, where each coalition is a subset of the set of all features. In other words, the SHAP value of a feature represents the change in the expected output of the model when that feature is included, taking into account all possible combinations of features (Lundberg et al., 2018).

$$\phi_j = \frac{1}{M} \sum_{S \subseteq X \setminus \{j\}} \frac{|S|!(|X|-|S|-1)!}{|X|!} [f(S \cup j) - f(S)] \quad (9)$$

where ϕ_j represents the SHAP value corresponding to feature j , while X denotes the entire set of features. The total number of potential feature coalitions is denoted by $f(S)$, and S refers to a subset of X that excludes feature j . The model's output when features $S \cup j$ are incorporated is represented by $f(S \cup j)$, whereas $f(S)$ signifies the model's output when only features S are considered.

2.4. Borehole data

To validate the optimized ML models on an independent dataset, borehole data, including measured conductivities, temperature gradients, and well-logging data, were utilized. The validation dataset was compiled from recent literature and covers sedimentary basins with diverse geological history in central, northern and eastern Europe. Data from seven boreholes, spanning depths between ca. 400 m and 4,000 m, were selected from the following locations: Danish Basin (DB; location: Aars; Denmark), North Germany Basin (NGB; locations: Ketzin, Adlershof, and Hannover; all Germany), Molasse Basin (MB; location: Bad Wurzach, Germany), and the Timan-Pechora Basin (TPB; exact locations: unknown, Russia).

Borehole data from the NGB and MB were introduced and extensively discussed in previous works (Fuchs et al., 2015). For this study, the dataset was extended with newly available borehole information. For the NGB, additional data were added from the shallow storage exploration well Berlin-Adlershof drilled in 2021 to a depth of ca. 456 m (cf. Norden et al., 2023). For the DB, borehole data were incorporated from the geothermal well Aars-1, which was drilled to a depth of 3.4 km in the Upper Keuper formation (Fuchs, 2018; Fuchs et al., 2020). We also used recent data for hydrocarbon wells from the Timan-Pechora Basin, which were reported by Meshalkin et al. (2020). The borehole datasets serve as independent validation benchmarks for assessing the prediction performance of ML models trained on synthetic data. A summary of borehole data used for model evaluation is provided in Table. 3.

Table 3. Laboratory TC datasets and petrophysical well-logs used for validation.

No.	Name	Basin	Borehole	Country	Formation	Type	Reference
1	Well A	NGB	Hannover	Germany	Triassic, Middle Buntsandstein, Volpriehausen Fm., Detfurth Fm.	Sandstone Marine playa	Fuchs et al. (2015)
2	Well B	NGB	Adlershof	Germany	Triassic, Exter Fm.	Sandstone Siltstone Claystone	Norden et al. (2023)
3	Well C	MB	Bad Wurzach	Germany	Obere Meereswassermolasse,	Limestone	Fuchs et al. (2015)

Micoene							
4	Well D	DB	Aars 1	Denmark	Triassic, Gassum Fm., Tertiary,	Mainly sandstone	Fuchs (2018) Fuchs et al. (2020)
5	Well E	TPB	Anonym, Well a	Russia	Carboniferous lower Permian	Mainly sandstone	Meshalkin et al. (2020)
6	Well F	TPB	Anonym, Well b	Russia	Carboniferous lower Permian	Mainly sandstone	Meshalkin et al. (2020)

3. Results

3.1. Synthetic data: predicted versus computed thermal properties

For each sedimentary rock group (clastic rocks, carbonates, and evaporites) and each target variable (TC, TD, and SHC), Fig.2 presents the MAPE of the best-performing model. The models were selected based on the lowest attainable MAPE across different numbers of well-log input combinations, ranging from single-log predictions to full four-log combinations. A total of 15 predictor combinations were evaluated for each rock group: four single-log models, six two-log combinations, four 3-log combinations, and one four-log combination. Across all evaluated models, XGBoost consistently achieved the lowest MAPE values, outperforming multiple linear regression, AdaBoost, and Random Forest (cf. Fig. 2). The critical number of well logs required to reach a sufficiently accurate prediction varies depending on the rock type, the target variable, and the applied statistical model. For thermal conductivity in clastic rocks, the majority of variance is explained with two input logs, with additional logs providing only minor improvement in predictive accuracy. In contrast, for carbonates and evaporates the inclusion of a third or fourth log significantly enhances predictive accuracy for TC. For TD and SHC, a similar trend was observed, although SHC prediction in clastic rocks showed a notable exception. Here, when using three or four log combinations, the linear model slightly outperformed the AdaBoost, albeit with levels of uncertainty.

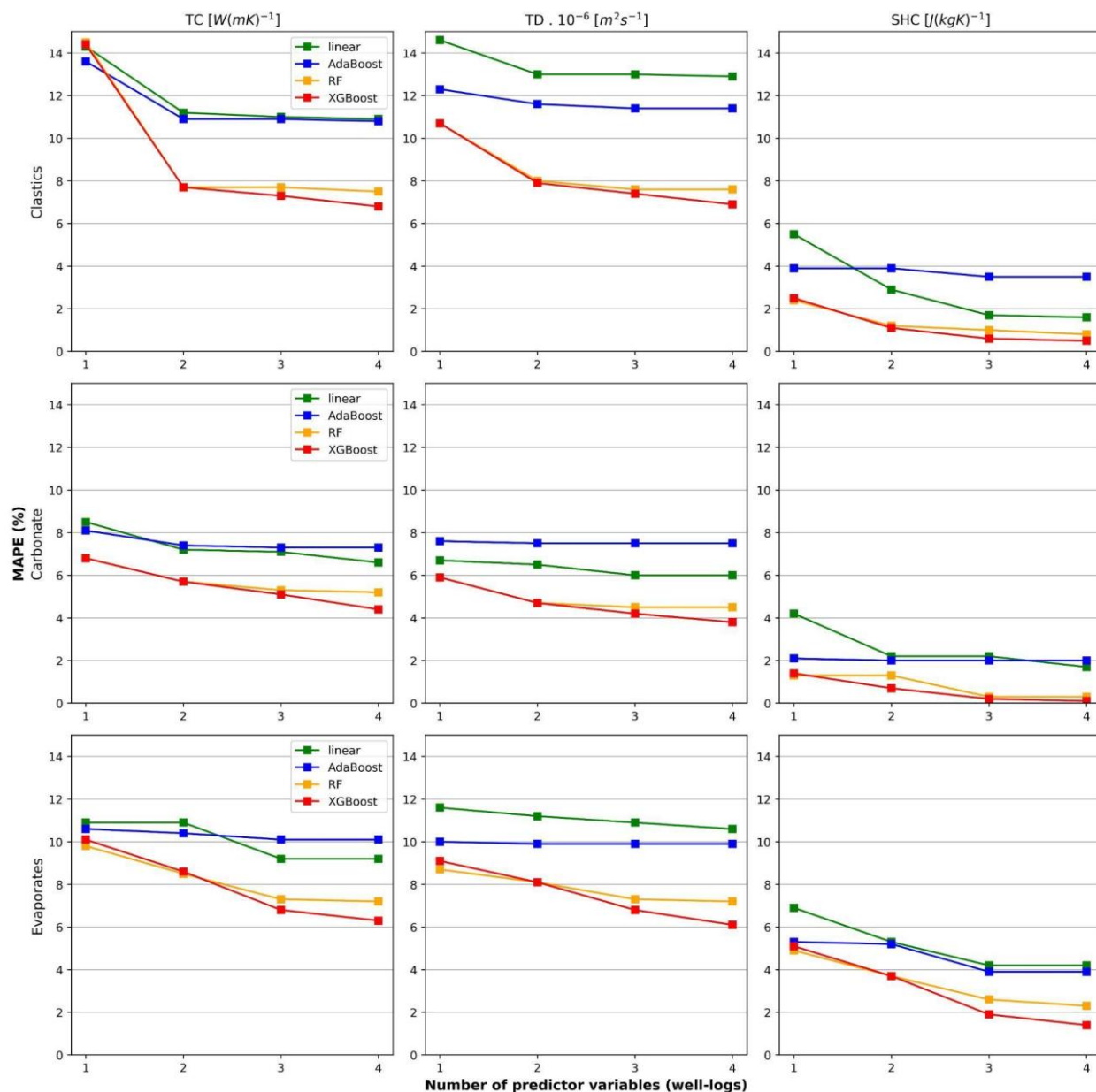


Fig. 2. Lowest mean absolute percentage error for predicting TC, TD, and SHC (columns) for each sedimentary rock group (rows) based on input features (x-axis). Prediction results for linear, AdaBoost, RF and XGBoost models are marked with green, blue, orange and red colors, respectively.

In Fig.3, the MAPE performance of the XGBoost are presented separately for all log combinations across the three rock groups (clastic rocks in green, carbonates in blue, and evaporates in orange). The following trends were observed (cf. also Appendices A, B, C for model weights and detailed statistics). In clastic rocks, for example, minimum MAPE of 7.7% (TC), 7.9% (TD), and 2.2% (SHC), and maximum MAPE of 13.1%, 13.4%, and 8.5% are observed when combining any two logs, respectively. In this rock group, for each of the target parameters, additional log information seems to further reduce the predictive accuracy of the less-preferable log combinations of high MAPE. Three-log combinations resulted in MAPEs below 10% for TC and TD, and below 5% for SHC in all cases. For carbonates, the

maximum MAPE is consistently below 8% for TC and TD and below 6% for SHC across all log combinations. For evaporates, combinations including PHIN perform significantly better and produce lower MAPE than combinations without PHIN (independent t-test, $p = <0.001$). The same accounts for TD and, to a less pronounced extent, for SHC. The worst stand-alone predictor variable in all cases is VSH.

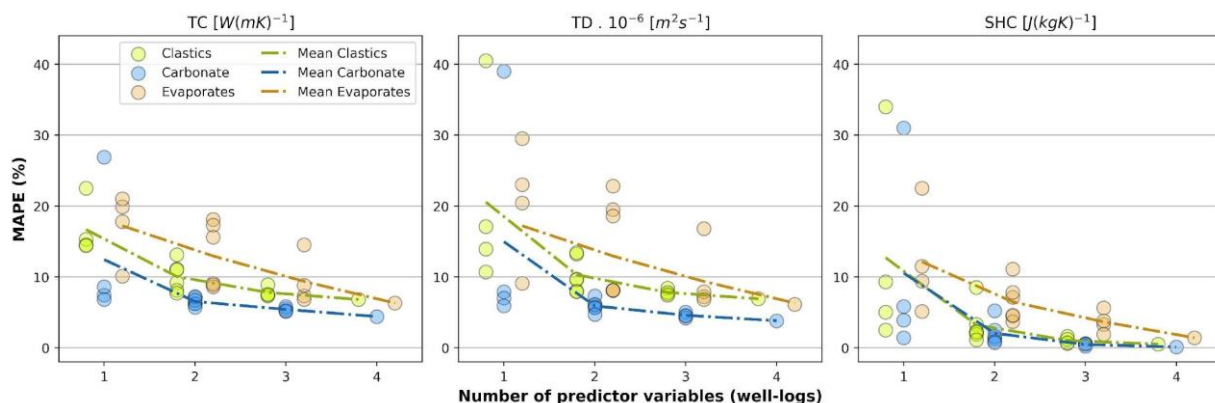


Fig. 3. XGBoost model performance (mean absolute percentage error MAPE) for different well-log combinations, ranging from log to four-log modes. The three panels represent predictions for TC, TD, and SHC. Results show different main sedimentary rock groups (green for clastic rocks, blue for carbonates, and orange for evaporates.). Dashed lines indicate the mean MAPE performance for each rock group across all log combinations.

The SHAP plot (see Fig. 4) is used to assess the influence of individual input features (logs as a predictor variable) for the predicted thermal properties (TC, TD, and SHC) as the target variable. The vertical axis ranks the predictor variables by importance, with the most influential appearing at the top and their impact decreasing downward. A key advantage of SHAP analysis is its ability to assign both positive and negative feature importance scores. Positive scores indicate that a given feature contributes to an increase in the predicted thermal property, while negative scores suggest the opposite. This allows for a clear distinction between features that are positively or negatively correlated with the target variable (Lundberg et al., 2020). The SHAP plot is particularly useful in identifying which features drive the model's predictions and whether their effects vary across different rock groups.

In Fig. 4., feature values are color-coded, with low values in blue and high values in red. The horizontal axis represents their effect on the model output, making it possible to observe how increasing or decreasing a feature influences predictions. For example, in clastic rocks, PHIN exhibits an inverse relationship with TC, meaning that as PHIN decreases, the predicted TC value increases as well. This suggests that lower porosity - represented by PHIN - correlates with higher thermal conductivity, which is consistent with known petrophysical behavior.

On the other hand, the importance of different logs decreases from top to bottom for predicting TC, indicating that PHIN has the highest impact for clastic rocks and evaporates, while it has the lowest impact for carbonates. *For heat*

capacity, we observe that VP has the largest impact across all rock groups and significantly reduces the model predictive error (independent t-test, $p = 0.002$).

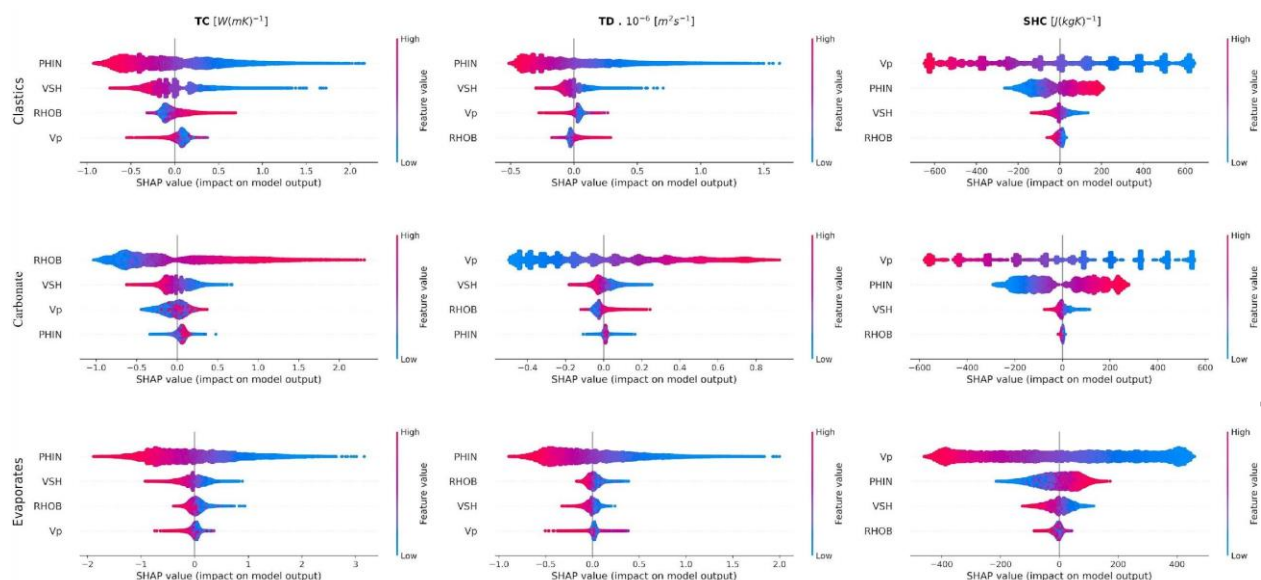


Fig. 4. SHAP features importance for synthetic sedimentary rock groups (clastic rocks, carbonates and evaporates). The x-axis shows feature impact on model predictions. The y-axis shows feature importance, with top features having the most impact and bottom features having the least. The colors represent the feature values, with red indicating higher values (High) and blue indicating lower values (Low) for each feature.

The non-monotonic relationship observed between feature values and SHAP values in Fig. 4, particularly for RHOB and Vp, reflects the nonlinear interactions between rock properties and the model's predictions. In geological systems, parameters such as bulk density and sonic velocity are governed by multiple interdependent factors, including porosity and mineral composition, leading to non-linear dependencies. The fluctuations seen in feature densities, such as Vp (Clastics-SHC, Carbonates-TD, and Carbonates-SHC), reflect a consequence of the sampling process. Specifically, the dataset was systematically designed by modifying matrix properties and incrementing porosity in 5% steps, which resulted in distinct patterns in SHAP values. The presence of 11 bubbles in carbonate and clastic rocks corresponds to the 11 porosity levels (ranging from 0% to 50% in 5% increments), indicating that these variations reflect underlying sub-patterns inherent in the dataset. The model's sensitivity to these sub-patterns also shows its ability to detect thresholds in the dataset, linking SHAP behavior to both data structure and subsurface physics.

The SHAP analysis further aids in interpreting geological models by revealing how features like porosity, shale volume, density, and sonic velocity influence predicted thermal property predictions (TC, TD, and SHC). For instance, higher porosity reduces thermal conductivity in clastic rocks, as increased void space impedes heat transfer. Similarly, shale volume affects thermal conductivity, given that shale typically exhibits lower conductivity than other rock types. Understanding these feature impacts enhances reservoir characterization, oil recovery strategies, and engineering applications by connecting machine learning predictions to real-world geological insights.

3.1.1. Thermal conductivity

The prediction accuracy of the models generally improves as the number of predictor variables increases (cf. Figures 2 and 3). For instance, in linear regression models, the MAPE values for clastic rocks decrease from 14.3% ($R^2=0.65$) with one predictor variable to 10.9% ($R^2=0.82$) with four predictor variables. A similar trend is observed for carbonates, where MAPE values drop from $8.5 \pm 6.6\%$ to $6.6 \pm 5.0\%$, and for evaporates, where the best XGBoost model achieves MAPE between 6.8 and 10.0% for any possible log combination. For carbonates with three predictor well-logs, the best prediction is achieved with XGBoost (A28-XGBoost; ame: $5.1 \pm 4.1\%$).

In general, the different regression models show varying levels of prediction accuracy. Comparing the mean uncertainties for different log-combinations and across the three rock groups, the linear and AdaBoost model performance clusters slightly outperformed by a cluster of RF and XGBoost models. The difference between these two model clusters is a drop in MAPE values by on average 2% and is of stronger impact for clastic rocks than for evaporates or carbonates. The maximum performance improvement using XGBoost instead of linear regression model is observed with ca. 5% absolute reduction of MAPE value for the combination of RHOB, PHIN, and VP in clastic rocks. Over all rock groups, the advantage of XGBoost compared to linear regression increases with the number of logs and thus an increasing complexity of the input data. The better performance of the ML models indicates that the relationship between the predictor variables and the target variables is not necessarily linear. Using multiple predictor variables generally improves the prediction accuracy of the test prediction models. For all rock groups and across all models, VSH is considered the worst single predictor variable for TC.

Fig.4 revealed the relative importance of different geological features on the TC model output for different rock groups. The results show that for the clastic rocks and evaporates, the most important features were PHIN, VSH, RHOB, and, Vp, in that order. For the carbonates group, the most important features were RHOB, VSH, Vp, and PHIN, in that order.

3.1.2. Thermal diffusivity and specific heat capacity

The fundamental observations noted for TC can be found similarly for TD and SHC. The maximum performance improvement using XGBoost instead of a linear regression model is observed for clastic rocks with an absolute reduction of MAPE value of 6.5 % for the combinations of RHOB, PHIN, and VP or PHIN and VSH, respectively. Over all log combinations, the lowest MAPE is observed for carbonates (7.8%), clastic rocks (12.1%), and evaporates (14.1%) with XGBoost regression. In general, the prediction of TD in carbonates performs significantly better than in the other rock groups (independent t-test, $p<0.001$; when excluding VSH only outlier). The advantage of XGBoost regression over linear regression becomes larger, the more logs are used as input variables. It has its maximum effect of around 5% for clastic rocks and averages to ca. 3% of MAPE drops for carbonates and evaporates (paired t-test, for all rock groups: $p<0.001$). Like for TC, the performance of the models can vary depending on the specific dataset and features used. VSH is still considered as the worst single predictor variable for all rock groups and across all models.

For SHC, the best prediction is achieved in carbonates and clastic rocks. Ignoring single VSH, which is by far the worst single predictor (MAPE > 30%), the MAPE values in both groups are averaging <3%. For evaporites the MAPE

for all log combinations average to 7%. For all three groups, the most important features were Vp, PHIN, VSH, and RHOB, in that order.

3.2. Evaluation on borehole data: predicted versus measured thermal conductivity

For the full dataset, the smallest prediction deviation is obtained by applying A41-XGBoost (including RHOB, PHIN, and VSH), which allows TC predictions with a mean deviation of $13.3 \pm 11.5\%$. Fig.5 presents histograms depicting the lowest deviations (best models) between measured and predicted values. This visualization is organized for each configuration of well logs, ranging from a minimum of one to a maximum of four, and various models including Linear, AdaBoost, RF, and XGBoost, encompassing the entire dataset.

Examining the green-colored plots on the left side of Fig.5, which represent the best linear model for different log combinations, the average error decreases from 19.6% to 17.5% as the number of predictor logs increases. However, for RF and XGBoost, the optimal model (with three logs=A41 and four logs=A45) demonstrated better performance when utilizing a combination of three logs rather than four logs. This could be because Vp has the least impact on the clastic rocks group compared to other logs in TC prediction.

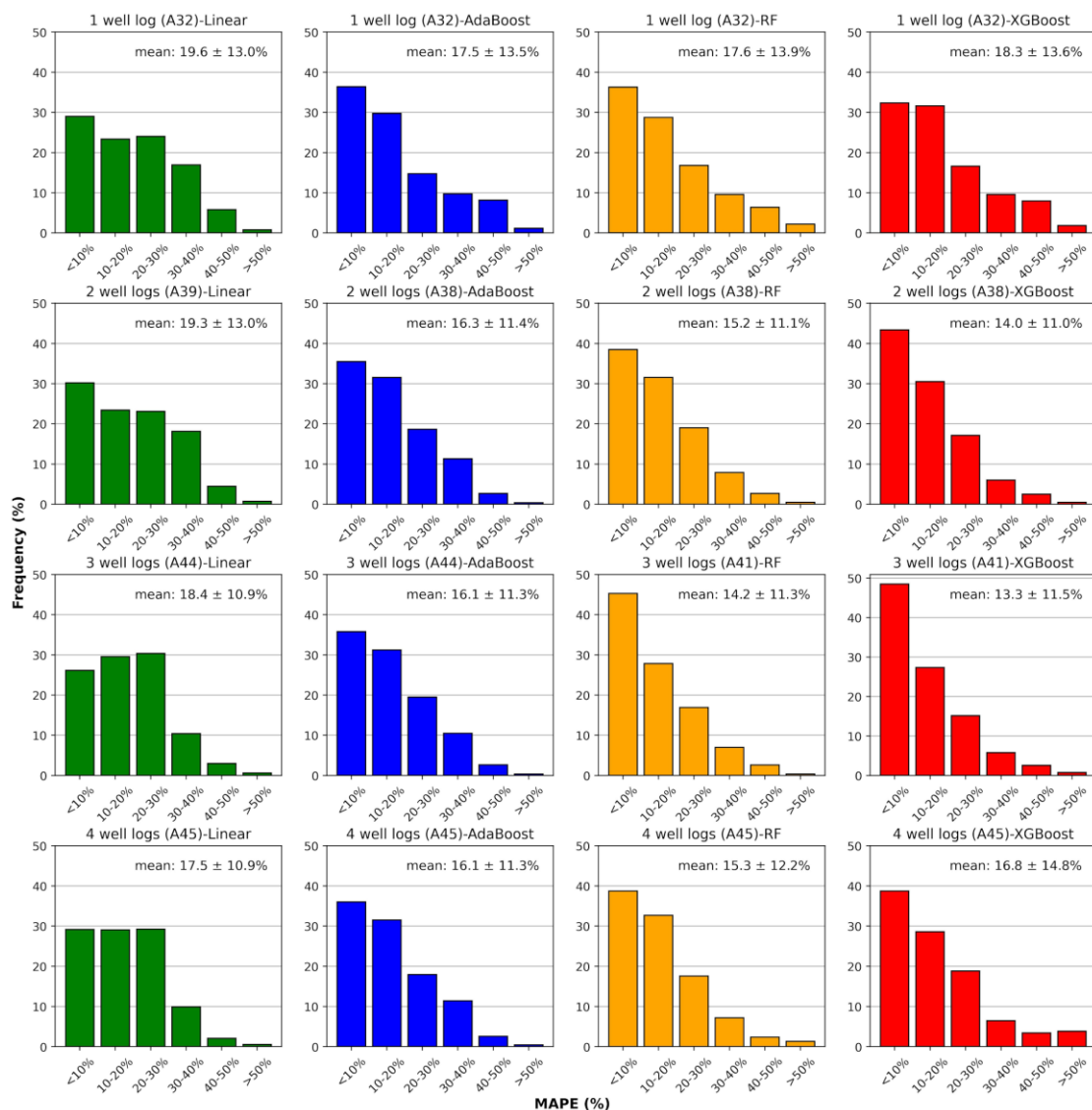


Fig. 5. Error histogram of residuals (predicted compared to measured TC) on full dataset for the highest-performance model with the lowest MAPE. Prediction results for linear, AdaBoost, RF, and XGBoost models are marked with green, blue, orange and red colors, respectively.

Fig.6 presents a scatter plot depicting the best model for each log combination based on the lowest MAPE values. Several key observations emerge from this analysis. First, both RF and XGBoost produce a more symmetrical distribution around the $y=x$ line compared to linear regression and AdaBoost, indicating a better overall fit. Second, XGBoost demonstrates superior performance in predicting TC values within the 4 - 5 W/mK range. Thirdly, in the prediction of TC in the combination of three well-logs, A41-RF and A41-XGBoost have achieved a lower-error value than the best linear model (A44-Linear).

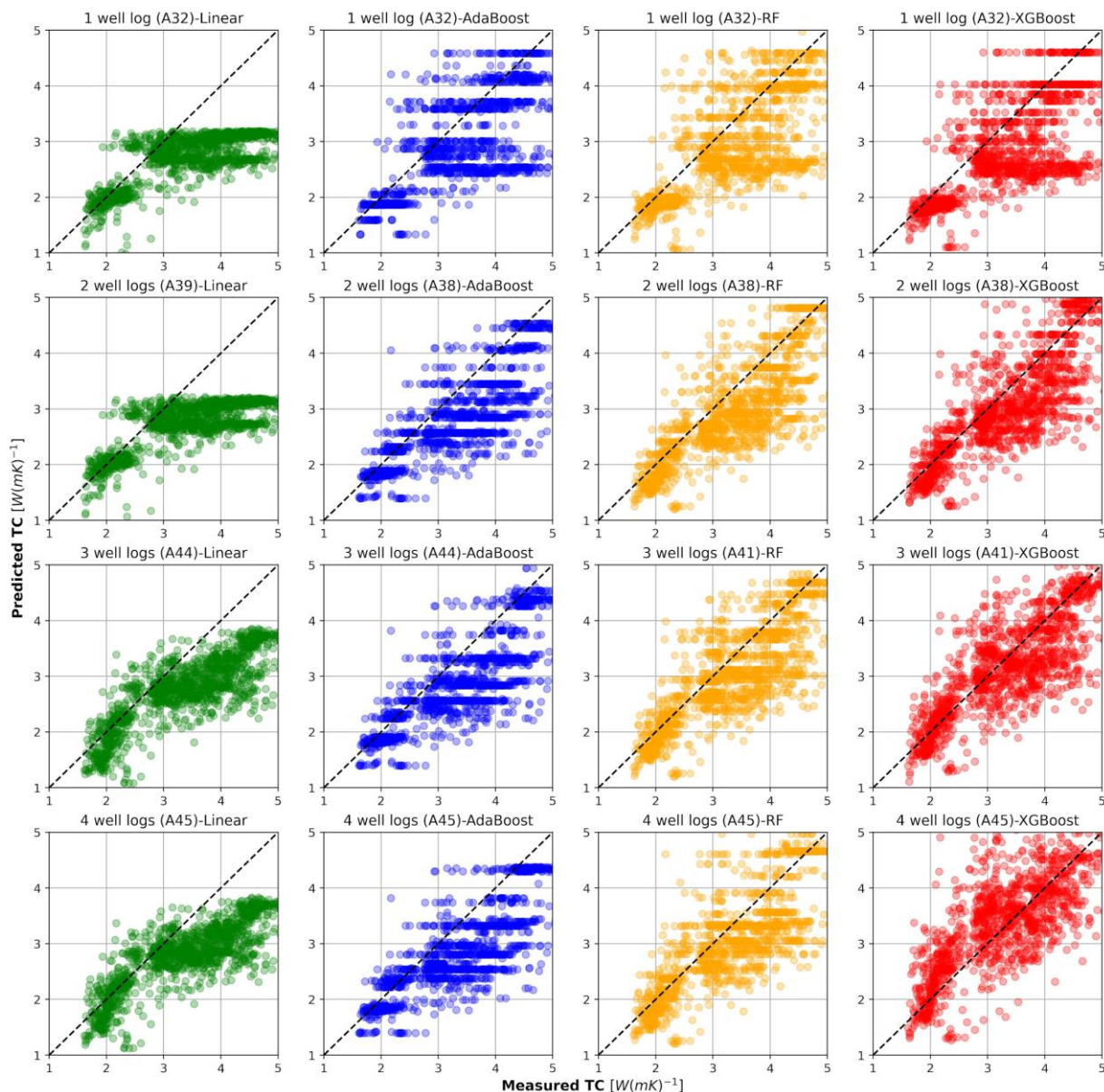


Fig. 6. The scatter plot between measured and predicted TC on the full dataset for the highest-performance model with the lowest MAPE. Prediction results for linear, AdaBoost, RF, and XGBoost models are marked with green, blue, orange and red colors, respectively.

The discrete stripe pattern in AdaBoost and RF predictions arises from their reliance on decision tree-based architectures, which create discrete steps by partitioning feature spaces into finite regions, leading to distinct horizontal bands in the scatter plots. These models assign predictions within discrete steps, as decision trees classify data by recursively splitting the input space into a series of fixed thresholds. In contrast, XGBoost mitigates this effect by leveraging regularized boosting techniques, which enables it to model continuous variations more effectively as more input features are introduced. By integrating gradient boosting and regularization, XGBoost refines tree splits dynamically, reducing overfitting and allowing for a more adaptive approximation of the underlying data distribution. As a result, XGBoost produces smoother prediction outputs, capturing subtler relationships within the dataset and

avoiding the rigid, step-like patterns seen in other tree-based models.

For geological formation intervals in the six wells (A to F), [Table 4](#) compares the well-log-based thermal conductivity predictions with thickness-weighted measured data from the same depth intervals. The intervals in meters and measured values with their standard deviations are listed, alongside the predicted values, residuals, and errors for the best models.

Table 4. Comparative analysis of thermal conductivity measured and predicted across different best-performing models for depth interval in six wells.

Name	Region	Interval (m)	Measured $\pm \text{std} (\frac{W}{mK})$	Model	Predicted $\pm \text{std} (\frac{W}{mK})$	Residual	Error (%)
Well A	NGB	3547 – 3563	3.93 ± 0.8	Linear	3.27 ± 0.4	0.66 ± 0.4	16.8
				AdaBoost	3.66 ± 0.3	0.27 ± 0.5	6.9
				RF	3.83 ± 0.9	0.10 ± 0.1	2.5
				XGBoost	3.98 ± 0.9	0.05 ± 0.1	1.3
Well B	NGB	214 – 454	2.39 ± 0.7	Linear	1.98 ± 0.3	0.41 ± 0.4	17.2
				AdaBoost	2.13 ± 0.3	0.26 ± 0.4	10.9
				RF	2.20 ± 0.3	0.19 ± 0.4	7.9
				XGBoost	2.26 ± 0.3	0.13 ± 0.4	5.4
Well C	MB	621 – 704	2.13 ± 0.3	Linear	1.98 ± 0.4	0.15 ± 0.1	7.0
				AdaBoost	1.98 ± 0.4	0.15 ± 0.1	7.0
				RF	2.01 ± 0.4	0.12 ± 0.1	5.6
				XGBoost	2.04 ± 0.4	0.09 ± 0.1	4.2
Well D	DB	1465 – 3345	2.16 ± 0.6	Linear	2.26 ± 0.5	0.10 ± 0.1	4.6
				AdaBoost	2.20 ± 0.4	0.04 ± 0.2	1.9
				RF	2.20 ± 0.5	0.04 ± 0.1	1.9
				XGBoost	2.25 ± 0.4	0.09 ± 0.2	4.2
Well E	TPB	1384 – 1513	2.51 ± 0.4	Linear	2.67 ± 0.3	0.16 ± 0.1	6.4
				AdaBoost	2.67 ± 0.3	0.16 ± 0.1	6.4
				RF	2.67 ± 0.6	0.16 ± 0.2	6.4
				XGBoost	2.63 ± 0.4	0.12 ± 0.0	4.8
Well F	TPB	1252 – 1559	2.63 ± 0.3	Linear	2.78 ± 0.4	0.15 ± 0.1	5.7
				AdaBoost	2.44 ± 0.4	0.19 ± 0.1	7.2
				RF	2.79 ± 0.4	0.16 ± 0.1	6.1
				XGBoost	2.76 ± 0.3	0.13 ± 0.0	4.9

The predictive capability of the derived models depends on the type and number of well logs used as well as the chosen ML approach. The most-effective well-log combination (RHOB, PHIN, and VSH) enable TC predictions with a MAPE of approximately 13% across the entire dataset. A substantial number of the models effectively replicate observed variations in TC across geological sections, as illustrated in Fig.7. Different log combinations, i.e. A31 to A45, are displayed for various models in gray. The A38 model (using PHIN and VSH), reveals that linear regression struggles to predict TC values in the range of 4 and 5 W/(m·K), whereas RF and XGBoost (particularly A36, A38, and A41) demonstrate better performance in capturing these values.

However, certain XGBoost-based models (A31, A33, and A34, and A37, which combine RHOB and Vp; cf. Appendix A), exhibit higher prediction errors, with an average MAPE of 26% and a standard deviation of 4%. These models perform less effectively in predicting TC values between 4 and 5 W/(m·K) compared to other log combinations.

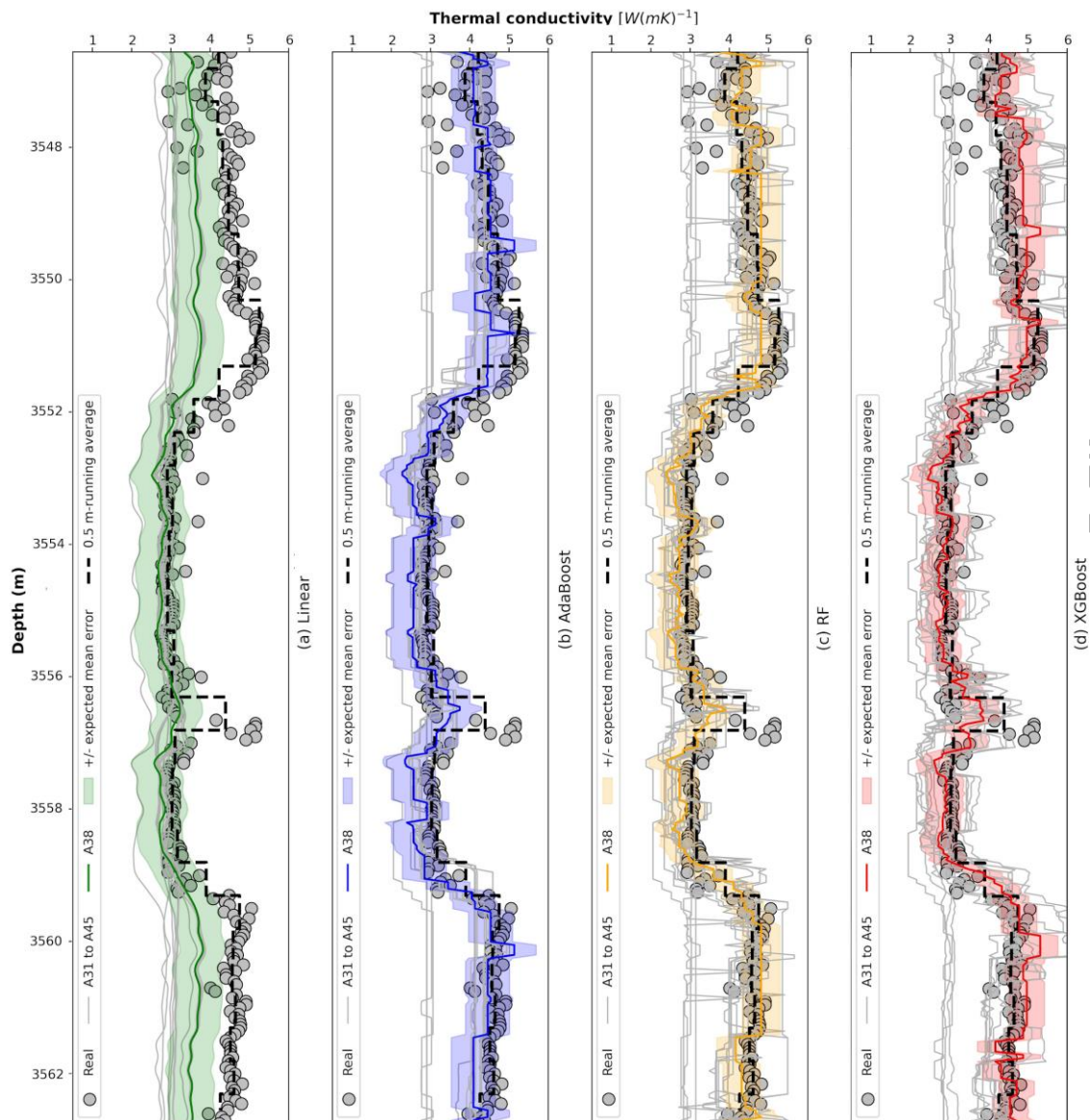


Fig. 7. All models (A31 to A45) have been illustrated to predict TC in well A. Models are depicted in gray, and the top-performing model (A41) is identified for linear, AdaBoost, RF, and XGBoost with colors in green, blue, orange and red, respectively.

Figure 8 displays scatter plots between predicted and measured thermal conductivity (TC) for different models across wells A to F. The corresponding histograms and error distributions are presented in Fig. 9

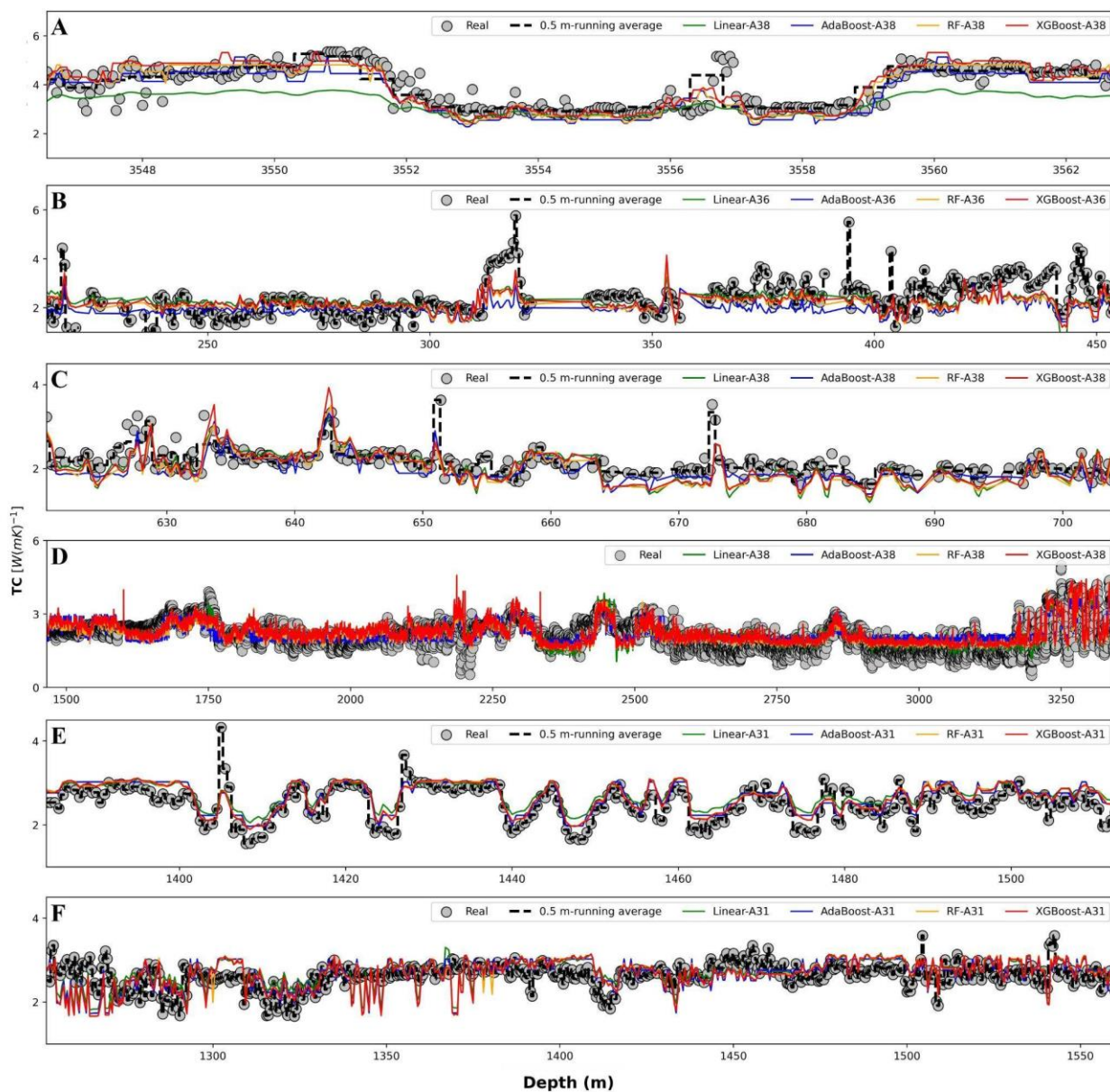


Fig. 8. Depth-dependent thermal conductivity (TC) profiles for wells A to F, comparing real measured values (black circles) with predicted values from different models (Linear, AdaBoost, RF, and XGBoost). The dashed black line represents a 0.5 m running average of the real TC values, while the colored lines correspond to different prediction models.

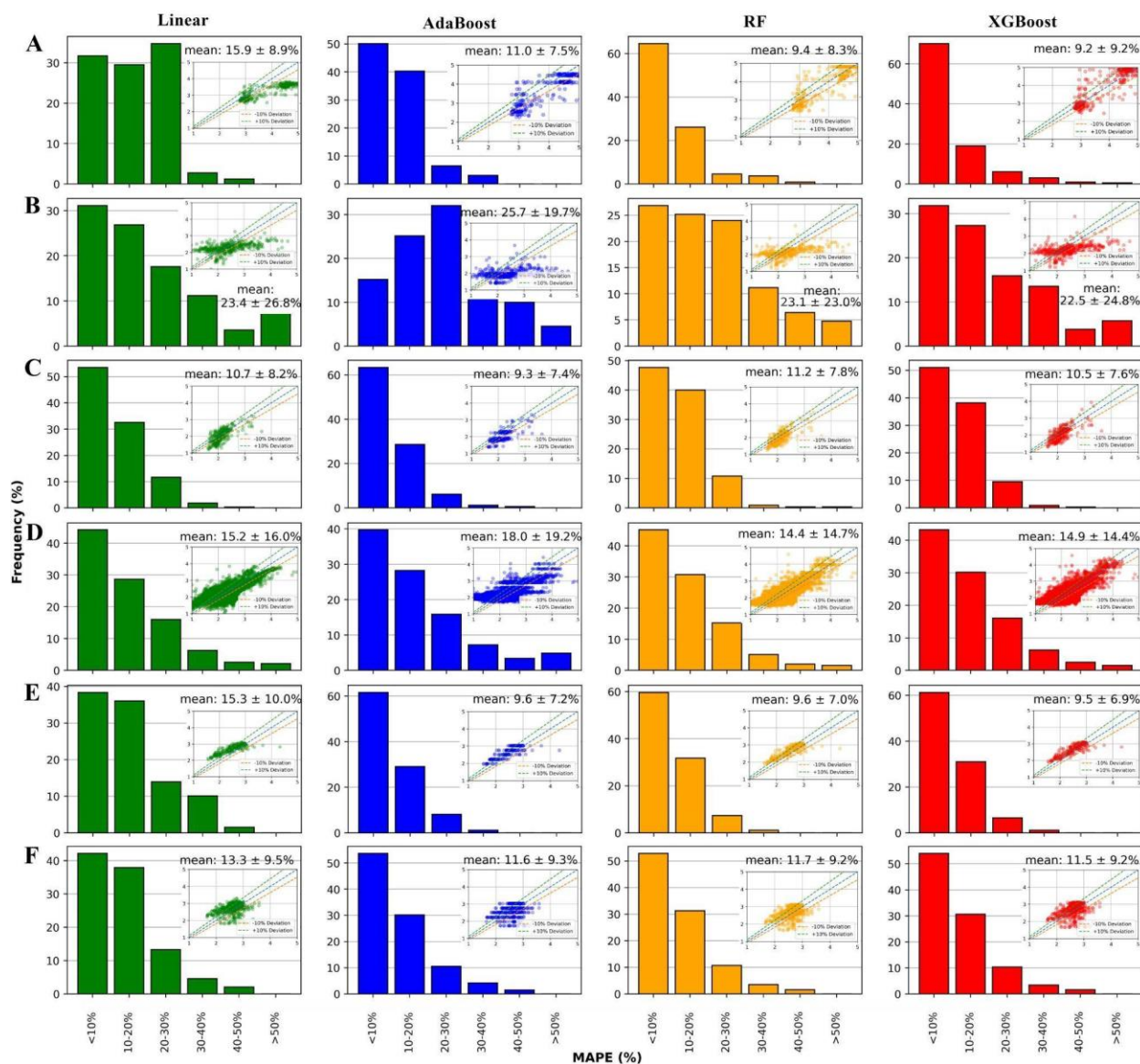


Fig. 9. Histogram plots showing the distribution of Mean Absolute Percentage Error (MAPE) for different models (Linear, AdaBoost, RF, and XGBoost) across wells A to F. Each histogram represents the frequency of errors within different percentage ranges. The embedded scatter plots illustrate the relationship between predicted and measured thermal-conductivity (TC) values for each model and well.

The RF and XGBoost models demonstrate superior predictive accuracy, with the majority of error distributions falling within the <10% and 10–20% ranges. In well A (Hannover Genesys, NGB), the linear model yields an average error of 15.9%, whereas XGBoost reduces this to 9.2%, marking a significant 6.7% decrease in MAPE (paired t-test, $p < 0.001$). In well B (Berlin-Adlershof, NGB), XGBoost performs slightly, but statistically insignificantly better ($p = 0.103$) than the linear model, achieving 0.9% reduction in MAPE.

In well C (A38, Bad Wurzbach, MB), the TC values for laboratory data, linear regression, AdaBoost, RF, and XGBoost are 2.13, 1.98, 1.98, 2.01, and 2.04 W/(m·K), respectively. Here, differences between model performances are insignificant ($p > 0.15$). In well D (A41, Aars, DB), the corresponding values are 2.16, 2.26, 2.20, 2.20, and 2.25 W/(m·K); again, the differences between the model predictions are insignificant (MAPE linear model: 10.7%, XGBoost: 10.5%). In well D, AdaBoost shows a significantly higher MAPE compared to the other models. RF and XGBoost yield slight non-significant MAPE reductions of 0.8% and 0.3% ($p > 0.25$), respectively, compared to the linear model.

The performance of the best well-log combination (A31) is assessed in well E (Anonym well a, Russia) and well F (Anonym well b, Russia). In well E, the differences between AdaBoost, RF, and XGBoost are insignificant (MAPE ca. 9.5%), but all together significantly better (all $p < 0.01$) than the linear model, exhibiting a notably high error rate. In well F, AdaBoost, RF, and XGBoost provide comparable results, with errors of approximately 11.5%. Additionally, RF and XGBoost show a higher frequency of errors below 10%, highlighting their improved predictive accuracy over the linear model.

The Fig. 10 illustrates the error percentage of different models and well-log combinations across wells A to F. The models compared include linear regression, Random Forest (RF), AdaBoost, and XGBoost, each represented by different symbols (circles, squares, triangles, etc.). The y-axis represents the error percentage, while the x-axis denotes the individual wells. Each model's performance is evaluated using various well-log combinations, ranging from single-log predictions to four-log models, indicated by different colors and size. The figure is divided into three colored regions: green (low error, below 20%), orange (medium error, between 20-30%), and white (high error, above 30%), providing a visual threshold for acceptable performance.

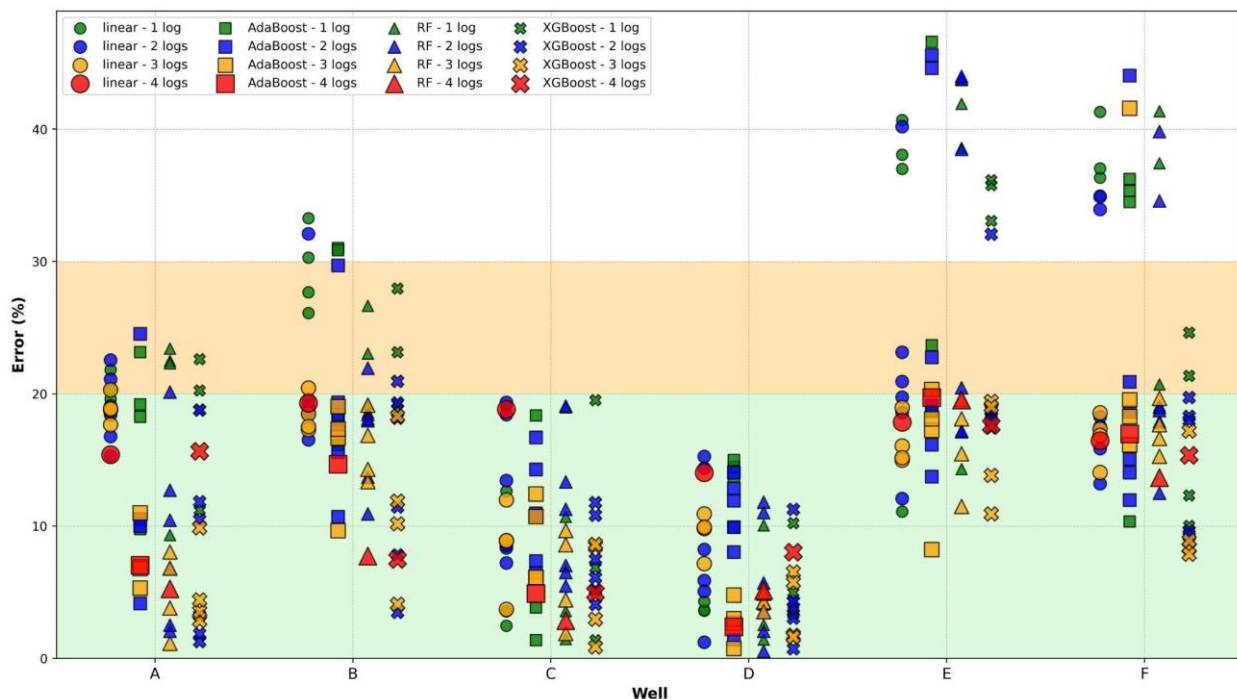


Fig. 10. Comparison of error percentages of thermal conductivity for various machine learning models (Linear, RF, AdaBoost, and XGBoost) across six wells (A–F) using different log combinations (1 to 4 logs).

As shown in Fig 10, error rates vary across models, log combinations, and wells. RF and XGBoost consistently perform slightly better than AdaBoost or linear regression. For RF and XGBoost, errors predominantly fall in the green region (<20%), particularly when three or four logs are used as predictor variables. The XGBoost model demonstrates strong predictive stability across most wells, maintaining low error margins across different geological settings. When using only one or two logs, the errors become more variable, with some models showing higher errors, particularly in wells E and F. This suggests that with fewer input variables, the models may lack sufficient data to make accurate predictions, leading to higher variability and occasional extreme errors. In contrast, when using 3 or 4 logs, errors generally appear more stable and concentrated in the lower error range. This observation was also made during validation on the synthetic test dataset. It is important to note that the errors reported in Fig. 11 apply to the scale of the predicted log data (0.5-m-scale) and the scale of the measured rock samples (cm scale). The effective errors on the formation scales are significantly smaller (cf. discussion section for further analysis).

Due to the lack of direct TD and SHC measurements, an indirect evaluation method was applied to assess these parameters. According to the heat equation, thermal conductivity (λ), thermal diffusivity (α), and the volumetric heat capacity (ρc) of a material are related to each other by (Fuchs et al., 2021):

$$\alpha = \frac{\lambda}{\rho c} \quad (10)$$

For well E density data were available. We used this well as an example for this cross-parameter evaluation (Fig. 11) TC, TD, and SHC profiles were predicted with the same combination of available well logs (A38, B38, and C38) and used as an input to compute the apparent density from equation 10. In the right track of Fig. 11, the logged and

predicted densities are plotted and compared.

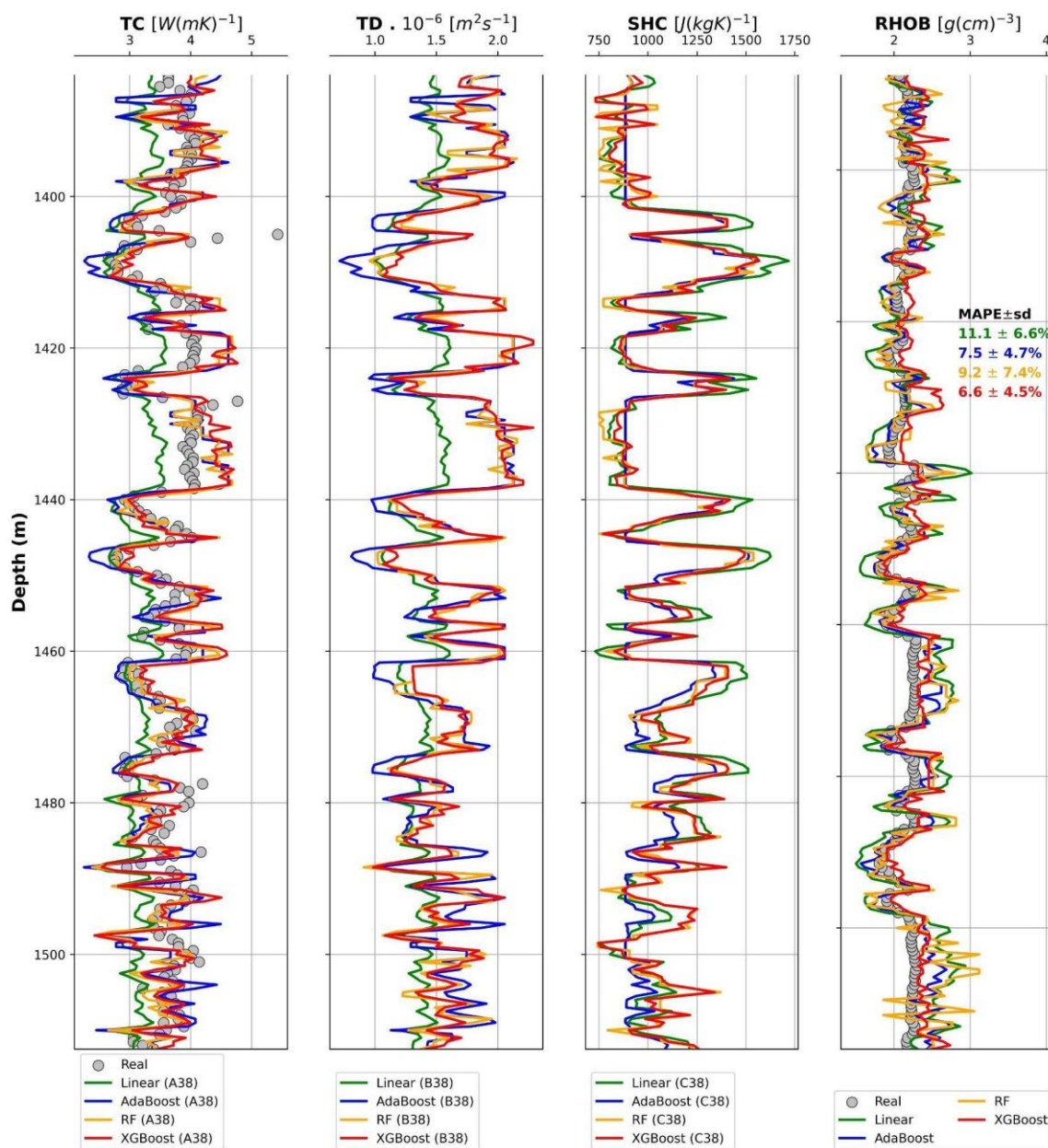


Fig. 11. Indirect cross-evaluation of predicted thermal properties (TC, TD, SHC) with the help of RHOB on the example of well E.

The model's validity is considered satisfactory if the measured bulk density values fall within the calculated bulk density range plus/minus the estimated uncertainty propagation. While this validation method demonstrates positive outcomes, additional validation using laboratory-measured TD and SHC data is necessary to further confirm the machine learning model's precision. For the linear regression, AdaBoost, RF, and XGBoost, the average errors along with their standard deviations are $11.1 \pm 6.6\%$, $7.5 \pm 4.7\%$, $9.2 \pm 7.4\%$, and $6.6 \pm 4.5\%$, respectively. XGBoost consistently achieves significantly lower error margins, reinforcing its effectiveness in predicting thermal properties.

Finally, we compare selected previously published empirical equations for TC with our approach. For an extended overview and literature references, we refer to. Fuchs & Förster (2014). The selection is based on the available logs in our data set and the application ranges reported in the original studies behind the traditional prediction equations. Table 5 presents a comparison of TC predictions using these previously published equations applied to our real wellbore data (validation dataset). The traditional equations exhibit errors exceeding 25%, whereas models presented in this study achieved significantly lower errors between 15 and 20%.

Table 5. Performance of previously published equations for thermal conductivity (TC) prediction computed on the validation wellbore dataset. The calculation based on the 0.5-m-resolution of the input well logs - a moving average was applied to TC point data to account for the well-log resolution.

Author (Year)	Equation	Lithotypes	MAPE (%)	Std (%)
Gross and Combs (1976)	$\lambda = 0.842 - 3.987\phi_N + 0.695V_p$	SS (cemented), SiS, GW	25.6	13.7
	$\lambda = -4.9\phi_N - 0.16V_p + 3.6\rho_b - 5.5$	SS, SiS, LI, DOL	36.5	23.5
Evans (1977)	$\lambda = -0.845 - 2.91\left(\frac{1}{V_p}\right) + 1.8\rho_b - 3.23V_{sh} + 1.714\left(\frac{1}{\phi_N}\right)^2$	SS	38.7	17.4
Hartmann et al. (2005)	$\lambda = -0.22 + 1.11\phi_N + 0.243V_p + 0.913\rho_b$	SS, DOL, LI	24.5	12.6
This study	model A39 (PHIN/VP)			
	Linear	all	19.3	14.1
	AdaBoost		17.1	12.2
	RF		17.0	13.0
	XGBoost		17.0	9.0
	model A42 (PHIN/VP/RHOB)			
	Linear	all	19.1	11.5
	AdaBoost		17.5	13.0
	RF		17.2	10.1
	XGBoost		18.6	9.4
	model A45 (PHIN/VP/RHOB/VSH)			
	Linear	all	17.5	12.1
	AdaBoost		16.1	13.5
	RF		15.3	15.1
	XGBoost		16.8	13.4

All of our presented ML models, based only on the training on synthetic datasets, significantly reduce TC prediction errors in clastic rocks compared to the applied traditional prediction equations, as evidenced by the lower MAPE values. It nicely demonstrates that the use of a generalized synthetic dataset and ML trained on these data produces

predictive models of better accuracy, which are not limited to the characteristics of an empirical dataset.

To evaluate the additional potential of explainable variance in the real data yet unexplained by the models trained on synthetic data, we implemented the wellbore data in a test training run of the pre-trained model (XGBoost-A45). This “calibration”, allows to adjust to well-specific characteristics. In this run, the pre-trained model was fine-tuned using the first 30% of each well's data in sequential order, while the remaining 70% was used for testing (not randomly split). In Table 6, the model's performance was then compared before and after calibration to determine how well it adapted to well-specific geological characteristics.

Table 6. Comparison of XGBoost-A45 before and after calibration using wellbore data.

Well name	MAPE (%) before calibration	MAPE (%) after calibration
Well A	15.1	11.3
Well B	9.7	8.4
Well C	8.4	6.1
Well D	9.1	6.7
Well E	18.5	11.4
Well F	16.4	12.1

The results in Table 6 indicate a consistent and statistically significant reduction in MAPE across all wells after calibration, demonstrating the effectiveness of calibration in improving model accuracy. For all wells, the MAPE reduced by absolute values ranging between 1.3 (well B) and 7.1%. The corresponding relative improvement is between 15 and 40%, which suggest that the fine-tuning process successfully adapted the model to well-specific geological characteristics, reducing prediction errors. The degree of improvement varied across wells, potentially reflecting differences in geological complexity, data quality, or the initial accuracy of the pre-trained model. However, this is only a test run to estimate the hidden variance in the real data, yet unexploited by our models trained on synthetic data. One key feature of our study is the general applicability to all sedimentary rocks for which site or well-bore-specific calibration would be counterproductive. Therefore, calibration was not performed on our general models and thus does not affect their performance statistics. This last test case is also not implemented in the open code base linked in the final part of this paper.

4. Discussion

General predictive capability

For variable input well-log combinations (A31–A45) and all tested models (see Appendices), the predictive capability for TC is illustrated in Figure 12. A lower MAPE value indicates a higher predictive accuracy, as demonstrated using a selected validation dataset. Among all models, XGBoost consistently produces the smallest predictive uncertainties. Certain well-log combinations result in higher MAPE values, indicating lower predictive accuracy. A33 (VSH), A34 (Vp), and A37 (RHOB, Vp) exhibit the highest MAPE, whereas the best-performing

combinations, like (RHOB, VSH), (PHIN, VSH), and (RHOB, PHIN, VSH) achieve the lowest MAPE (~13%). The well logs used in this analysis reflect different petrophysical properties: PHIN (hydrogen index) and less pronounced RHOB (density) and VP (sonic velocity) primarily serve as porosity indicators, VSH may reflect the clay mineral content, assuming that the gamma-ray index reflects the radiating effect of clay rather the perturbing effects like mud.

Combining multiple well logs generally enhances predictive performance compared to using single-log models. All two-log and multi-log combinations, except for A37 (RHOB, Vp), yield MAPE values below 20% at the log scale (0.5 m). Notably, pairing porosity indicators (PHIN, RHOB, or Vp) with a lithotype indicator (VSH) significantly improves predictions, whereas VSH alone is a less accurate predictor. Using three or four well logs does not necessarily lead to a substantial reduction in MAPE but can help stabilize predictions, reducing variability in model outputs. For instance, the A42 and A43 combinations, which incorporate RHOB and Vp (combined in A37) with an additional log, achieve predictive uncertainties comparable to other three-log models, suggesting that the weaker performance of specific two-log combinations can be mitigated by additional well-log data.

The results presented (Figures 7–13; Tables 4–5) highlight the superiority of the trained ML models, with XGBoost consistently outperforming AdaBoost and RF (which Meschalkin et al., 2020 was in favor for), and always exceeding the accuracy of linear regression models – both those developed in this study and those proposed by Fuchs et al. (2015) – at both log and formation scales. While the absolute increase in prediction accuracy varies, the improvement is statistically significant across all model comparisons, as confirmed by variance analysis of group means.

Validation and comparison with previously published equations

The validation of the trained models in this study is based on (i) real borehole data with measured TC values, (ii) indirect cross-parameter correlations between TC, TD, SHC, and RHOB, and (iii) comparisons with established empirical approaches from the literature. While thermal conductivity (TC) in clastic rocks benefits from a strong validation dataset compiled from previous studies (Wells A–F), available data for carbonates, evaporites, thermal diffusivity (TD), and specific heat capacity (SHC) remain scarce. While TC data for wells A to D were available in digital format, the data for wells E and F were digitized from published figures (Meshalkin et al., 2020; Fig. 3 and 4). This digitization process may have introduced random uncertainties in both the log data and the target TC values, potentially affecting the accuracy of the extracted data. Previous studies (Fuchs & Förster, 2014; Fuchs et al., 2015) already demonstrated the advantage of synthetic datasets over regionally and geologically limited borehole-derived datasets and their associated empirical prediction equations (cf. Demongodin et al., 1991; Hartmann et al., 2005; Hu et al., 2021). More recent efforts have trained AI models on such limited datasets (e.g., Gainitdinov et al., 2024; Meshalkin et al., 2020; Sharo et al., 2022), achieving better prediction accuracy than traditional linear regression. However, these models often lack generalizability across different sedimentary environments.

Our approach combines large synthetic datasets derived primarily from physical laws (logged parameters and SHC) and empirical models (TC, TD), with ML-based statistical techniques. These models capture non-linear interdependencies far more effectively than conventional linear regression approaches. As a result, our trained models are not constrained by regional limitations, making them applicable across diverse sedimentary formations and adaptable to various well-log datasets. To ensure a reasonable and fair comparison, we selected multiple-linear

regression equations from the literature that utilize at least two or three well-log components (cf. Table 5). The original authors reported prediction errors of $\pm 10\%$ ($0.3 \text{ W/(m}\cdot\text{K)}$, Goss & Combs, 1976), $\pm 15\%$ (Evans, 1977), and $\pm 10\%$ ($0.3 \text{ W/(m}\cdot\text{K)}$, Hartmann et al., 2005). However, applying these empirical equations to our borehole dataset resulted in significantly higher average uncertainties (MAPE): 25% and 36% (Goss & Combs, 1976), 38% (Evans, 1977), and 25% (Hartmann et al., 2005). In contrast, our ML models, trained on generalized synthetic datasets, consistently produced lower MAPE values (15–20%) for the same well-log combinations. This confirms that ML-based predictions trained on synthetic datasets yield superior accuracy, independent of region-specific empirical datasets. Among the tested models, XGBoost consistently delivers the highest predictive performance, reinforcing its universal applicability to sedimentary rocks. However, our borehole datasets used for empirical validation just covers continental sedimentary basins with porosities likely between 0 and 30%. The evaluation of predicted TC for sedimentary rocks of larger porosity, incorporating values between 30 and 50% typically for shallow sediments in marine settings, is still open.

Scale effect on interval TC predictions

The errors reported in this study reflect differences between measured rock sample data (centimeter scale) and predicted well-log data (0.5-meter scale). However, for numerical modeling applications, geological formations are typically represented by effective average thermal properties over much larger interval scales (tens to hundreds of meters). For calculations of heat flow density, intervals for average thermal conductivity can reach some thousand meters, for example, when surface temperature and a single bottom-hole temperature (BHT) at typical hydrocarbon exploration depth of 3 to 5 km are used for a gradient calculation. This is where the predictive capability of the presented machine-learning models becomes particularly valuable, as they allow for the computation of interval-based thermal properties rather than laboratory-scale values (Table 4; Figure 14). For the six independent borehole datasets analyzed, with interval lengths ranging from 16 to 1,880 meters, the interval thermal conductivity (TC) was computed using a thickness-weighted harmonic mean based on the measured TC values. The best-performing model, XGBoost, achieved errors between 1.3% and 5.4% (equal to less than $0.15 \text{ W/(m}\cdot\text{K)}$ in each case). Similar or slightly higher errors were observed for RF and AdaBoost, whereas linear regression performed well for longer intervals but exhibited larger errors for shorter intervals ($<100 \text{ m}$), particularly in wells A and C. The lowest interval-based errors ($\sim 5\%$) at scales of 10–100 meters are substantially smaller than the point-based MAPE errors (5–20%) at centimeter to half-meter scales. This reduction in error is statistically significant (independent t-test; all $p < 0.001$) and primarily due to averaging effects, where local over- and underestimations at the point scale tend to cancel each other out over longer intervals. However, the reliability of this effect depends on the absence of systematic bias in the prediction models. Given the significant error reduction with increasing interval length and the consistent performance of XGBoost, there is no indication of systematic over- or underestimation bias in the model. For heat-flow calculations and geothermal modeling, this capability is particularly beneficial. Reliable average TC values can be computed for entire borehole intervals in deep boreholes, complementing single-point measurements such as bottom-hole temperature or formation test data. This approach effectively expands the availability of high-quality heat-flow data, improving data accuracy and completeness in global databases such as the IHFC Global Heat Flow Database (Fuchs et al., 2021; 2023, 2024).

Application ranges and methodological limitations

The presented prediction approaches rely on the generation of large synthetic datasets for sedimentary rocks, serving as the computational foundation for training machine-learning (ML) models. While these synthetic datasets offer a broad and representative database, ML techniques inherently capture non-linear interrelations between standard geophysical logging data and thermal rock properties more effectively than conventional regression models (cf. Meshalkin et al., 2020). The synthetic dataset computation follows an adapted multi-mineral approach, based on two primary assumptions outlined by Fuchs et al. (2015). The first assumption is that the global compositional variability of sedimentary rocks from continental and marine basins is sufficiently represented by 15 major rock-forming minerals, as defined in Table 1 for the three primary rock groups (clastic rocks, carbonates, and evaporites). The second assumption is that well-log response equations, including physical laws and mixing models, are sufficiently understood to approximate the true thermal properties of any given multi-mineral rock composition. While the arithmetic mean model used for computing specific heat capacity (SHC) and bulk density (RHOB) is physically well-founded, the geometric mean model used for thermal conductivity (TC) is empirical rather than theoretical, yet it remains the best approximation for TC in sedimentary rocks. The reported model uncertainty for TC predictions falls within 5–10% (Fuchs et al., 2013), and similar uncertainties apply to thermal diffusivity (TD), which is calculated using a geometric mean model modified by an arithmetic mean component (Goto & Matsubayashi, 2009; Fuchs et al., 2021). Regarding the mineral end-member values used in the model calculations based on Fuchs et al. (2015), no significant updates have been introduced in the past decade in the literature, suggesting that relative uncertainties remain below 5%. A third assumption is that ML models (AdaBoost, RF, XGBoost) and linear regression techniques are appropriate for exploring the complex interrelations between petrophysical and thermal parameters.

Traditional linear and multilinear equations often struggle to capture the complex relationships between geological parameters, particularly when different rock types exhibit distinct or non-linear dependencies. These limitations complicate accurate thermal property predictions, from which we know that they follow non-linear relations in sedimentary rocks. Machine learning can model these non-linear interactions and enhance predictive power by leveraging diverse algorithms and feature selection techniques, allowing models to generalize beyond the given dataset. This is reflected by the consistently better predictive accuracy of the presented XGBoost models (and slightly less the RV) compared to multiple linear regression analysis. AdaBoost did not provide an advantage compared to linear regression approaches on average.

The accuracy of the trained ML models when applied to borehole data depends not only on the accumulated uncertainties from data generation, model training, and underlying assumptions but also on the technical quality of geophysical well logs and the reliability of well-log interpretation. While model-related uncertainties remain constant for any user, the handling of input well-log data introduces additional sources of error. Careful and consistent log calibration and correction for environmental effects – such as logging speed, borehole diameter, mud type, tool positioning, casing effects, and mud cake thickness – are essential to minimize uncertainties (cf. Schlumberger, 2009). Key well logs used in this study include neutron porosity index (PHIN), bulk density (RHOB), sonic velocity (V_p), and gamma-ray (GR) logs. Notably, volume fraction of shale (VSH) is derived as gamma ray index from GR measurements without incorporating age, regional, or rock-type-specific corrections from the literature (e.g., Larionov,

1969; Stieber, 1970; Clavier et al., 1971). This follows the assumption that industry-standard log response values, as compiled in sources such as Serra (1984) and Schlumberger log interpretation charts, also do not apply these corrections. In cases where gamma-ray data is unavailable, a spontaneous potential (SP) log can serve as an alternative input for older boreholes.

For further methodological considerations, particularly regarding pressure and temperature effects, we refer to Fuchs et al. (2015). Their analysis suggests that DT and NPHI logs partially capture these effects, but they also recommend careful post-processing of predicted thermal property values using empirical, in-situ corrections (cf. Norden et al., 2020).

A key limitation of our approach is that the current models do not explicitly account for anisotropy effects, which can significantly influence thermal conductivity and diffusivity across different scales, particularly in layered or fractured sedimentary formations (cf. Chekhonin et al., 2018). The absence of comprehensive, high-resolution datasets currently prevents a detailed analysis of anisotropy and its relationship with thermal and well-log parameters. Future studies incorporating directional thermal property measurements and anisotropy-aware modeling techniques could provide deeper insights into these effects.

Open questions for future research

This study leaves room for future improvements. The current synthetic dataset is based on an empirical geometric mean model for thermal conductivity. This model could be refined or replaced once a more fundamental physical expression is derived that better explains observed conductivity data in sedimentary rocks. The same applies to the thermal-diffusivity (TD) model, which would change accordingly. Another limitation is the lack of direct borehole validation data for TD and specific heat capacity (SHC). Incorporating laboratory-measured TD and SHC values in future research would enhance validation efforts and improve model reliability. Moreover, measurement data that also cover samples with porosities between 30% and 50%, like in marine sediments, would further strengthen the validation efforts. Additionally, anisotropy effects, which can significantly impact TC and TD, particularly in layered or fractured formations, are not explicitly considered. An open question remains whether directional petrophysical and thermal property measurements could be integrated into a log-derived anisotropy model, improving accuracy in geologically complex settings. Further research should also explore the development of a generalized in-situ framework that accounts for pressure-temperature-fluid interactions affecting thermal properties. While some pT effects are indirectly captured by well-log data, additional corrections may be required, and a systematic approach to these adjustments would improve the predictive power of ML models. This study focuses exclusively on sedimentary rocks, yet many geothermal and hydrocarbon applications require knowledge of thermal properties in crystalline basement formations. Extending the method to non-sedimentary rocks could benefit from transfer learning, where a model pre-trained on sedimentary rock data adapts to new geological domains with limited available data. This approach could also be useful for improving generalizability between different sedimentary rock groups.

While this study demonstrated the effectiveness of tree-based ensemble models, particularly XGBoost, due to their interpretability and computational efficiency, future work could explore deep learning (DL) techniques, such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), or Transformers, to further improve thermal property prediction. DL models may uncover deeper feature relationships within the data, enhancing accuracy,

especially for complex or spatially correlated well-log datasets. However, their implementation would require larger datasets, significant computational resources, and careful architectural optimization. Investigating DL approaches could not only boost predictive performance but also improve generalizability across different sedimentary rock groups, extending the framework developed in this study.

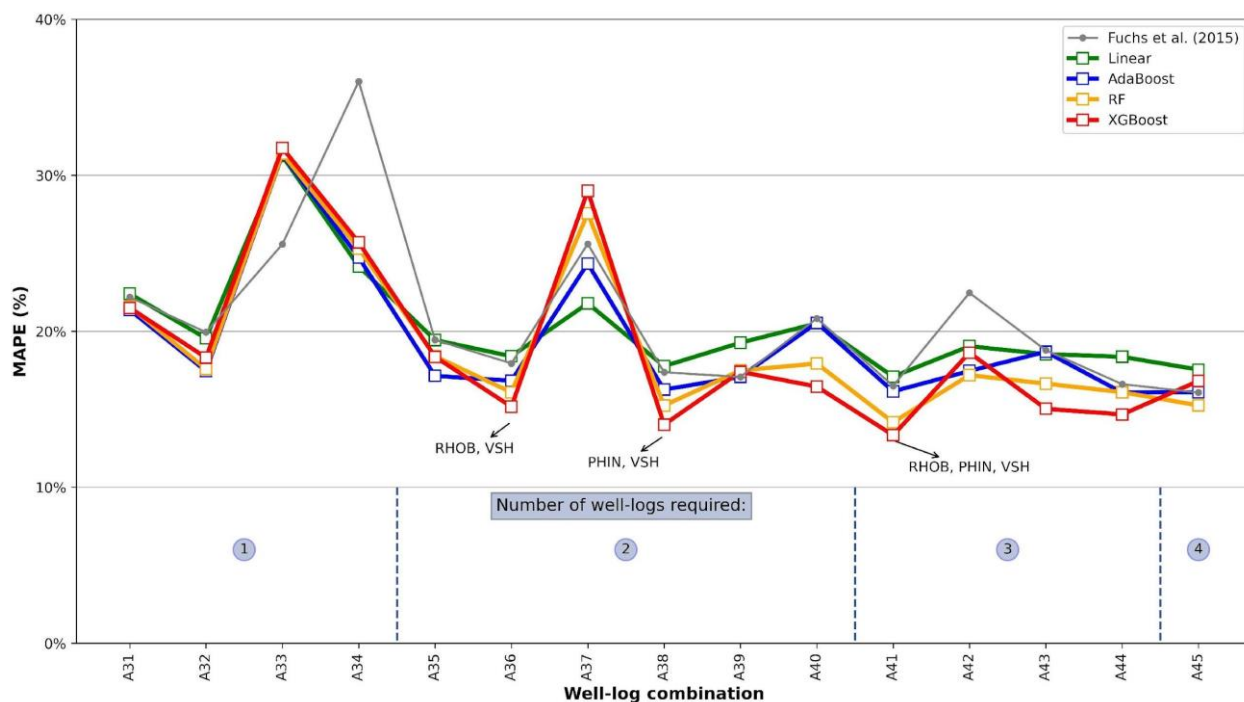


Fig. 12. The mean absolute percentage error of predicting thermal conductivity using four regression models (Linear, AdaBoost, RF, and XGBoost) with different well-log combinations was calculated using a real validation dataset for clastic rocks.

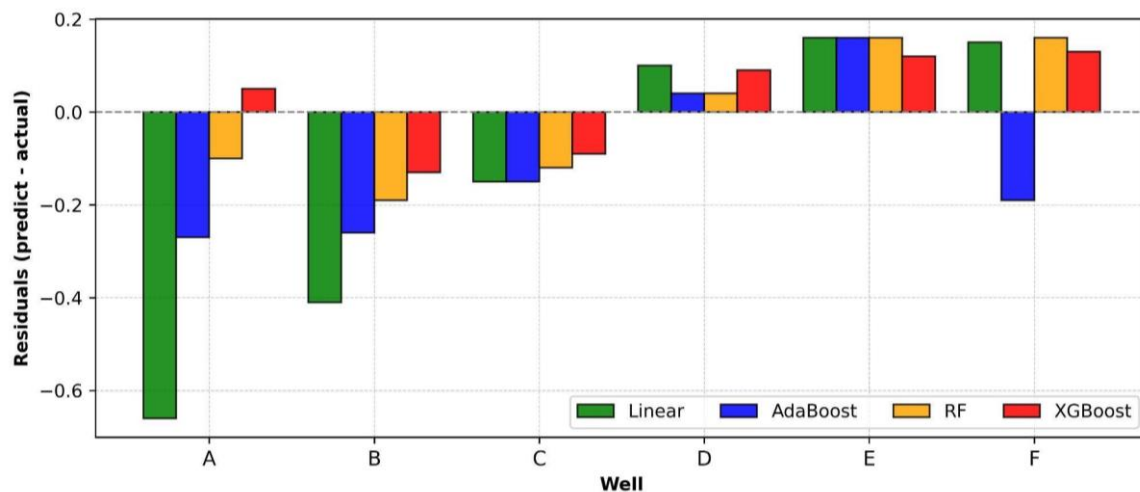


Fig 13: Overview of full well-predicted residual profiles (wells A to F) for best-performance models. Each well has residuals from four models: linear (green), AdaBoost (blue), RF (orange), and XGBoost (red).

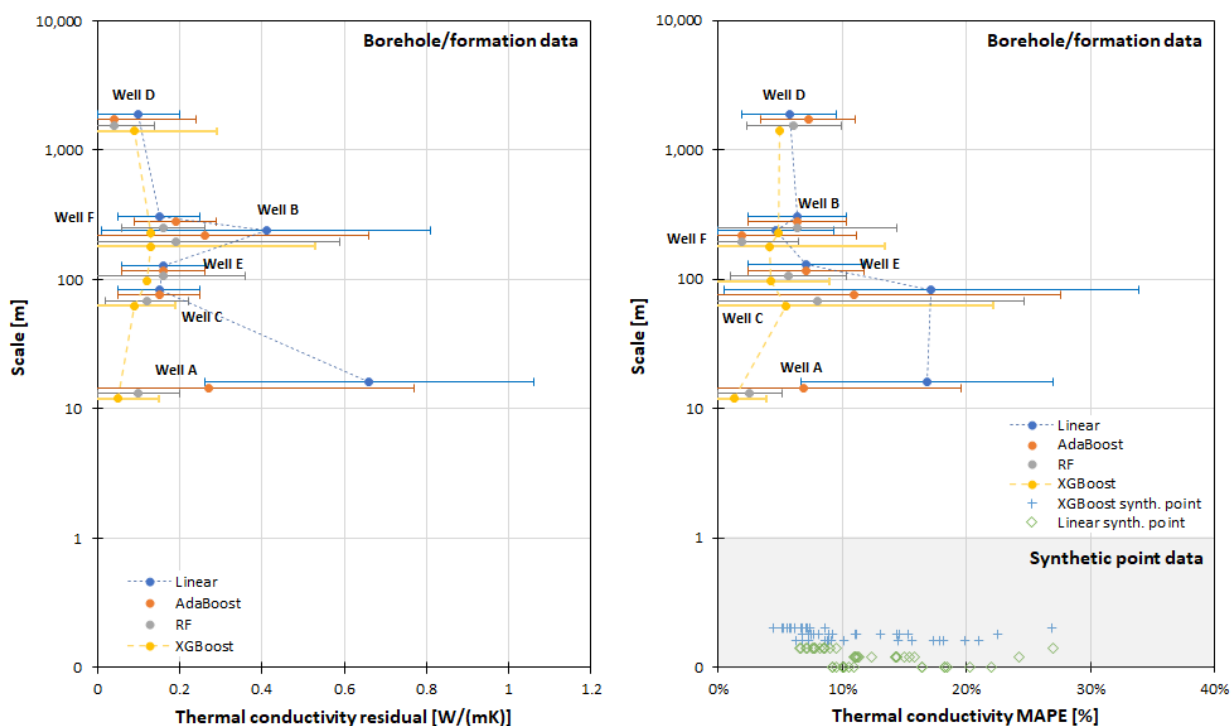


Fig 14: Impact of scale on thermal conductivity predictions across multiple wells (A to F), comparing methods (Linear, Adaboost, XGBoost, RF) with residual errors and MAPE.

5. Summary and conclusion

This study developed and validated machine learning (ML) models for predicting thermal conductivity (TC), thermal diffusivity (TD), and specific heat capacity (SHC) in sedimentary rocks based on standard geophysical well logs (bulk density, natural gamma-ray, sonic velocity, hydrogen index, and petrophysical descriptors derived from these). The models were trained on a large synthetic dataset, incorporating multi-mineral mixing models and physical well-log response equations. The results demonstrate that XGBoost frequently outperforms other models, including Random Forest (RF), AdaBoost, and linear regression, at both log scale (0.5 m) and formation scale (tens to hundreds of meters). The predictive accuracy improves with multi-log combinations, particularly when porosity indicators (PHIN, RHOB, Vp) are combined with lithology indicators (VSH). However, we report models trained for any possible log combination to make the application of our results more flexible. XGBoost achieves the lowest mean absolute percentage errors (MAPE) for predicting TC in boreholes on the log scale, typically between 15% and 20%, compared to higher errors (>25%) in empirical models applied from the literature. These empirical equations from literature struggle to generalize across different geological settings, while ML models trained on synthetic data offer broader applicability across diverse sedimentary formations. Interval-based predictions (formation to borehole scale) show reduced errors (interval mean errors ca. 5%), confirming that averaging effects mitigate local point-scale inaccuracies. Validation with real borehole data confirms that ML-based predictions outperform traditional regression models, though thermal diffusivity (TD) and specific heat capacity (SHC) lack direct borehole validation data.

Main conclusions are:

- This study confirms the superiority of ML approaches over traditional regression models for predicting thermal rock properties in sedimentary rocks from geophysical well logs. The XGBoost model trained on synthetic data emerges as the most reliable and recommended model, demonstrating low uncertainty across different sedimentary rock types. The use of synthetic datasets is the fundamental base that enables generalized predictions, eliminating regional biases found in models that are based on empirical borehole data.
- The ability to compute interval-based thermal properties on any geological formation or borehole scale with an uncertainty of ca. 5% or less, rather than relying on point-scale laboratory data, significantly enhances the parameterization of subsurface thermal models. Applying our trained tool to multiple boreholes in sedimentary basins will allow us to systematically assess litho-facies-dependent variations in thermal rock properties. This is particularly valuable for geothermal energy, hydrocarbon exploration, and reservoir modeling, and will serve improved heat-flow calculations with smaller uncertainties in the future.
- Despite the improvements in predictive accuracy, future research is needed to address anisotropy effects, pressure-temperature corrections, and the applicability of models to crystalline basement rocks. Additionally, incorporating deep learning techniques could further enhance pattern recognition and prediction accuracy in complex geological environments.

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Data availability

The open-source tool is available via <https://github.com/mojtaba355/Thermal-Properties-Calculation--AI> (First release: March. 2024; Python 3 with modules: numpy, pandas, joblib, xgboost, and scikit-learn).

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Appendix A. TC models for well-log combinations. R^2 , MAPE, SD, and RMSPE are calculated on the synthetic test dataset.

	Model	Well-logs	Coefficients	R^2	MAPE (%)	SD (%)	RMSPE (%)
Evaporites	A1-Linear	RHOB	$a_0 = -1.803, a_1 = 2.079$	0.194	20.3	15.2	25.3
	A1-AdaBoost	RHOB		0.199	20.1	15.1	25.1
	A1-RF	RHOB		0.210	19.9	15.1	24.9
	A1-XGBoost	RHOB		0.211	19.9	15.0	24.9
	A2-Linear	PHIN	$a_0 = 4.579, a_1 = -6.528$	0.745	10.9	1.0	24.9
	A2-AdaBoost	PHIN		0.768	10.6	8.3	13.4
	A2-RF	PHIN		0.793	9.8	7.8	12.5
	A2-XGBoost	PHIN		0.790	10.1	7.8	12.6
	A3-Linear	VSH	$a_0 = 3.484, a_1 = -1.926$	0.147	22.0	17.9	28.4
	A3-AdaBoost	VSH		0.148	22.0	17.9	28.4
	A3-RF	VSH		0.148	22.0	17.9	28.4
	A3-XGBoost	VSH		0.136	21.0	16.6	26.8
	A4-Linear	Vp	$a_0 = 0.312, a_1 = 0.653$	0.323	18.2	15.1	23.7
	A4-AdaBoost	Vp		0.338	18.1	15.1	23.6
	A4-RF	Vp		0.350	17.8	14.9	23.3
	A4-XGBoost	Vp		0.355	17.8	14.9	23.2
	A5-Linear	RHOB PHIN	$a_0 = 8.191, a_1 = -1.407, a_2 = -8.113$	0.793	10.1	8.1	13.0
	A5-AdaBoost	RHOB PHIN		0.797	10.4	8.2	13.2
	A5-RF	RHOB PHIN		0.827	9.2	7.3	11.7
	A5-XGBoost	RHOB PHIN		0.828	9.1	7.2	11.6
	A6-Linear	RHOB VSH	$a_0 = -1.088, a_1 = 1.984, a_2 = -1.806$	0.342	18.4	15.2	23.8
	A6-AdaBoost	RHOB VSH		0.281	19.0	15.4	24.4
	A6-RF	RHOB VSH		0.330	18.1	15.1	23.6
	A6-XGBoost	RHOB VSH		0.331	18.1	15.0	23.5
	A7-Linear	RHOB Vp	$a_0 = 1.825, a_1 = -1.063, a_2 = 0.879$	0.337	18.2	15.6	24.0
	A7-AdaBoost	RHOB Vp		0.346	18.1	15.1	23.5
	A7-RF	RHOB Vp		0.396	17.4	15.5	23.3
	A7-XGBoost	RHOB Vp		0.407	17.3	15.4	23.1
	A8-Linear	PHIN VSH	$a_0 = 4.800, a_1 = -6.193, a_2 = -1.105$	0.793	10.0	7.6	12.5
	A8-AdaBoost	PHIN VSH		0.804	10.1	7.8	12.5
	A8-RF	PHIN VSH		0.847	8.5	6.5	10.7
	A8-XGBoost	PHIN VSH		0.846	8.6	6.5	10.7
	A9-Linear	PHIN Vp	$a_0 = 6.389, a_1 = -8.301, a_2 = -0.338$	0.780	10.5	8.3	13.3
	A9-AdaBoost	PHIN Vp		0.798	10.4	8.3	13.3
	A9-RF	PHIN Vp		0.838	8.9	7.0	11.3

Carbonates	A9-XGBoost	PHIN Vp		0.838	8.9	6.9	11.2
	A10-Linear	VSH Vp	$a_0 = 0.921, a_1 = -1.657, a_2 = 0.615$	0.433	16.4	15.4	22.5
	A10-AdaBoost	VSH Vp		0.432	17.0	15.5	23.0
	A10-RF	VSH Vp		0.481	15.5	15.1	21.6
	A10-XGBoost	VSH Vp		0.476	15.6	15.1	21.7
	A11-Linear	RHOB PHIN VSH	$a_0 = 8.009, a_1 = -1.258, a_2 = -7.647, a_3 = -0.988$	0.831	9.2	7.1	11.7
	A11-AdaBoost	RHOB PHIN VSH		0.825	10.1	8.1	13.0
	A11-RF	RHOB PHIN VSH		0.887	7.5	5.8	9.5
	A11-XGBoost	RHOB PHIN VSH		0.893	7.3	5.7	9.3
	A12-Linear	RHOB PHIN Vp	$a_0 = 8.081, a_1 = -1.174, a_2 = -8.328, a_3 = -0.091$	0.795	10.1	8.1	13.0
	A12-AdaBoost	RHOB PHIN Vp		0.800	10.2	8.1	13.0
	A12-RF	RHOB PHIN Vp		0.838	8.9	7.0	11.3
	A12-XGBoost	RHOB PHIN Vp		0.841	8.8	7.0	11.1
	A13-Linear	RHOB VSH Vp	$a_0 = 2.146, a_1 = -0.867, a_2 = -1.628, a_3 = 0.801$	0.442	16.4	15.9	22.8
	A13-AdaBoost	RHOB VSH Vp		0.433	16.9	15.3	22.8
	A13-RF	RHOB VSH Vp		0.512	14.9	15.7	21.7
	A13-XGBoost	RHOB VSH Vp		0.542	14.5	15.7	21.3
	A14-Linear	PHIN VSH Vp	$a_0 = 6.397, a_1 = -7.799, a_2 = -1.024, a_3 = -0.301$	0.820	9.5	7.2	11.9
	A14-AdaBoost	PHIN VSH Vp		0.818	10.0	7.9	12.7
	A14-RF	PHIN VSH Vp		0.888	7.3	5.8	9.4
	A14-XGBoost	PHIN VSH Vp		0.901	6.8	5.5	8.7
	A15-Linear	RHOB PHIN VSH Vp	$a_0 = 7.910, a_1 = -1.050, a_2 = -7.841, a_3 = -0.986, a_4 = -0.082$	0.832	9.2	7.1	11.6
	A15-AdaBoost	RHOB PHIN VSH Vp		0.826	10.1	8.1	13.0
	A15-RF	RHOB PHIN VSH Vp		0.892	7.2	5.7	9.2
	A15-XGBoost	RHOB PHIN VSH Vp		0.916	6.3	5.1	8.2
	A16-Linear	RHOB	$a_0 = -3.951, a_1 = 2.801$	0.860	8.5	6.6	10.8
	A16-AdaBoost	RHOB		0.875	8.0	6.4	10.2
	A16-RF	RHOB		0.879	7.5	6.2	9.7
	A16-XGBoost	RHOB		0.879	7.4	6.2	9.7
	A17-Linear	PHIN	$a_0 = 3.855, a_1 = -4.472$	0.829	9.5	7.1	11.8
	A17-AdaBoost	PHIN		0.842	9.1	6.8	11.4
	A17-RF	PHIN		0.846	8.7	6.6	10.9
	A17-XGBoost	PHIN		0.845	8.6	6.5	10.8
	A18-Linear	VSH	$a_0 = 2.974, a_1 = -1.820$	0.080	27.0	18.7	32.9
	A18-AdaBoost	VSH		0.079	27.4	19.2	33.4
	A18-RF	VSH		0.080	27.0	18.7	32.9
	A18-XGBoost	VSH		0.080	26.9	18.5	32.6
	A19-Linear	Vp	$a_0 = -0.124, a_1 = 0.672$	0.855	8.6	6.3	10.7
	A19-AdaBoost	Vp		0.870	8.1	6.3	10.7
	A19-RF	Vp		0.889	6.8	5.6	8.8

A19-XGBoost	Vp		0.887	6.8	5.6	8.8
A20-Linear	RHOB PHIN	$a_0 = -2.573, a_1 = 2.311, a_2 = -0.820$	0.861	8.5	6.5	10.7
A20-AdaBoost	RHOB PHIN		0.882	7.9	6.2	10.0
A20-RF	RHOB PHIN		0.888	7.3	5.9	9.4
A20-XGBoost	RHOB PHIN		0.893	7.1	5.8	9.1
A21-Linear	RHOB VSH	$a_0 = -3.466, a_1 = 2.740, a_2 = -1.174$	0.892	7.8	6.0	9.9
A21-AdaBoost	RHOB VSH		0.908	7.4	5.7	9.4
A21-RF	RHOB VSH		0.920	6.3	4.9	7.9
A21-XGBoost	RHOB VSH		0.922	6.2	4.8	7.8
A22-Linear	RHOB Vp	$a_0 = -2.283, a_1 = 1.540, a_2 = 0.317$	0.876	7.7	6.0	9.7
A22-AdaBoost	RHOB Vp		0.884	7.8	6.1	9.9
A22-RF	RHOB Vp		0.905	6.3	5.4	8.3
A22-XGBoost	RHOB Vp		0.909	6.2	5.3	8.1
A23-Linear	PHIN VSH	$a_0 = 4.135, a_1 = -4.364, a_2 = -1.066$	0.856	9.0	6.7	11.2
A23-AdaBoost	PHIN VSH		0.865	8.8	6.5	11.0
A23-RF	PHIN VSH		0.890	7.2	5.7	9.2
A23-XGBoost	PHIN VSH		0.892	7.2	5.6	9.1
A24-Linear	PHIN Vp	$a_0 = 1.086, a_1 = -1.407, a_2 = 0.472$	0.861	8.2	6.2	10.3
A24-AdaBoost	PHIN Vp		0.870	8.1	6.1	10.1
A24-RF	PHIN Vp		0.889	6.7	5.7	8.8
A24-XGBoost	PHIN Vp		0.889	6.7	5.7	8.8
A25-Linear	VSH Vp	$a_0 = 0.323, a_1 = -1.364, a_2 = 0.660$	0.899	7.2	5.5	9.1
A25-AdaBoost	VSH Vp		0.897	7.6	5.7	9.5
A25-RF	VSH Vp		0.923	5.7	4.7	7.4
A25-XGBoost	VSH Vp		0.923	5.7	4.6	7.4
A26-Linear	RHOB PHIN VSH	$a_0 = -2.754, a_1 = 2.486, a_2 = -0.427, a_3 = -1.161$	0.893	7.8	6.0	9.9
A26-AdaBoost	RHOB PHIN VSH		0.907	7.4	5.6	9.3
A26-RF	RHOB PHIN VSH		0.926	5.9	4.6	7.5
A26-XGBoost	RHOB PHIN VSH		0.938	5.5	4.2	6.9
A27-Linear	RHOB PHIN Vp	$a_0 = -2.971, a_1 = 1.726, a_2 = 0.496, a_3 = 0.344$	0.876	7.6	6.0	9.7
A27-AdaBoost	RHOB PHIN Vp		0.885	7.8	6.0	9.9
A27-RF	RHOB PHIN Vp		0.912	6.1	5.3	8.0
A27-XGBoost	RHOB PHIN Vp		0.919	5.8	5.1	7.7
A28-Linear	RHOB VSH Vp	$a_0 = -1.504, a_1 = 1.281, a_2 = -1.265, a_3 = 0.365$	0.913	6.7	5.1	8.4
A28-AdaBoost	RHOB VSH Vp		0.912	7.2	5.4	9.0
A28-RF	RHOB VSH Vp		0.937	5.3	4.1	6.8
A28-XGBoost	RHOB VSH Vp		0.942	5.1	5.5	6.6
A29-Linear	PHIN VSH Vp	$a_0 = 0.851, a_1 = -0.632, a_2 = -1.316, a_3 = 0.571$	0.900	7.1	5.5	9.0
A29-AdaBoost	PHIN VSH Vp		0.898	7.6	5.7	9.5
A29-RF	PHIN VSH Vp		0.929	5.5	4.5	7.1

A29-XGBoost	PHIN VSH Vp		0.934	5.2	4.4	6.8
A30-Linear	RHOB PHIN VSH Vp	$a_0 = -3.289, a_1 = 1.761, a_2 = 1.316, a_3 = -1.327, a_4 = 0.442$	0.916	6.6	5.0	8.3
A30-AdaBoost	RHOB PHIN VSH Vp		0.911	7.3	5.5	9.1
A30-RF	RHOB PHIN VSH Vp		0.939	5.2	4.2	6.7
A30-XGBoost	RHOB PHIN VSH Vp		0.958	4.4	3.6	5.6
A31-Linear	RHOB	$a_0 = -2.621, a_1 = 2.075$	0.565	15.8	12.0	19.8
A31-AdaBoost	RHOB		0.565	15.7	12.0	19.7
A31-RF	RHOB		0.574	15.4	12.0	19.5
A31-XGBoost	RHOB		0.574	15.3	11.9	19.4
A32-Linear	PHIN	$a_0 = 3.187, a_1 = -3.617$	0.648	14.3	10.3	17.6
A32-AdaBoost	PHIN		0.701	13.6	9.8	16.8
A32-RF	PHIN		0.726	12.6	9.3	15.7
A32-XGBoost	PHIN		0.706	14.6	9.2	15.5
A33-Linear	VSH	$a_0 = 3.023, a_1 = -1.795$	0.279	24.2	17.4	29.8
A33-AdaBoost	VSH		0.289	24.0	17.2	29.5
A33-RF	VSH		0.291	23.9	17.1	29.4
A33-XGBoost	VSH		0.291	22.5	15.2	27.1
A34-Linear	Vp	$a_0 = -0.133, a_1 = 0.604$	0.588	15.4	10.8	18.8
A34-AdaBoost	Vp		0.594	15.3	10.9	18.8
A34-RF	Vp		0.615	14.5	10.8	18.1
A34-XGBoost	Vp		0.615	14.4	10.7	17.9
A35-Linear	RHOB PHIN	$a_0 = 2.824, a_1 = 0.134, a_2 = -3.417$	0.649	14.3	10.3	17.6
A35-AdaBoost	RHOB PHIN		0.692	13.6	9.8	16.8
A35-RF	RHOB PHIN		0.760	11.5	9.2	14.7
A35-XGBoost	RHOB PHIN		0.770	11.1	9.1	14.4
A36-Linear	RHOB VSH	$a_0 = -1.488, a_1 = 1.957, a_2 = -1.568$	0.771	12.4	9.8	15.8
A36-AdaBoost	RHOB VSH		0.815	11.7	8.8	14.6
A36-RF	RHOB VSH		0.853	9.3	7.4	11.9
A36-XGBoost	RHOB VSH		0.855	9.2	7.4	11.8
A37-Linear	RHOB Vp	$a_0 = -1.136, a_1 = 0.775, a_2 = 0.399$	0.600	15.0	10.9	18.5
A37-AdaBoost	RHOB Vp		0.589	15.1	11.0	18.7
A37-RF	RHOB Vp		0.663	13.4	10.8	17.2
A37-XGBoost	RHOB Vp		0.668	13.1	10.6	16.8
A38-Linear	PHIN VSH	$a_0 = 3.863, a_1 = -3.325, a_2 = -1.388$	0.806	11.2	9.1	14.4
A38-AdaBoost	PHIN VSH		0.831	10.9	7.8	13.4
A38-RF	PHIN VSH		0.876	8.1	6.8	10.6
A38-XGBoost	PHIN VSH		0.876	8.1	6.8	10.6
A39-Linear	PHIN Vp	$a_0 = 3.612, a_1 = -4.054, a_2 = -0.080$	0.649	14.4	10.3	17.7
A39-AdaBoost	PHIN Vp		0.693	13.6	9.8	16.7
A39-RF	PHIN Vp		0.769	11.0	9.6	14.6

A39-XGBoost	PHIN Vp		0.764	11.0	9.7	14.7
A40-Linear	VSH Vp	$a_0 = 0.854, a_1 = -1.579, a_2 = 0.573$	0.797	11.3	8.8	14.3
A40-AdaBoost	VSH Vp		0.813	11.4	8.3	14.1
A40-RF	VSH Vp		0.886	7.7	6.5	10.1
A40-XGBoost	VSH Vp		0.888	7.7	6.5	10.0
A41-Linear	RHOB PHIN VSH	$a_0 = 2.364, a_1 = 0.558, a_2 = -2.481, a_3 = -1.427$	0.813	11.0	9.0	14.2
A41-AdaBoost	RHOB PHIN VSH		0.834	10.9	7.8	13.3
A41-RF	RHOB PHIN VSH		0.884	7.8	6.5	10.1
A41-XGBoost	RHOB PHIN VSH		0.891	7.5	6.3	9.8
A42-Linear	RHOB PHIN Vp	$a_0 = 3.142, a_1 = 0.248, a_2 = -3.891, a_3 = -0.118$	0.650	14.4	10.2	17.7
A42-AdaBoost	RHOB PHIN Vp		0.703	13.6	9.7	16.7
A42-RF	RHOB PHIN Vp		0.819	9.5	8.6	12.8
A42-XGBoost	RHOB PHIN Vp		0.840	8.9	8.0	12.0
A43-Linear	RHOB VSH Vp	$a_0 = 0.001, a_1 = 0.654, a_2 = -1.568, a_3 = 0.400$	0.805	11.1	8.8	14.2
A43-AdaBoost	RHOB VSH Vp		0.823	11.3	8.2	13.9
A43-RF	RHOB VSH Vp		0.889	7.6	6.4	9.9
A43-XGBoost	RHOB VSH Vp		0.897	7.3	6.2	9.6
A44-Linear	PHIN VSH Vp	$a_0 = 2.721, a_1 = -2.105, a_2 = -1.455, a_3 = 0.220$	0.812	11.0	8.8	14.1
A44-AdaBoost	PHIN VSH Vp		0.834	10.9	7.8	13.4
A44-RF	PHIN VSH Vp		0.890	7.7	6.4	10.0
A44-XGBoost	PHIN VSH Vp		0.894	7.5	6.3	9.8
A45-Linear	RHOB PHIN VSH Vp	$a_0 = 1.927, a_1 = 0.415, a_2 = -1.820, a_3 = -1.465, a_4 = 0.158$	0.815	10.9	8.8	14.0
A45-AdaBoost	RHOB PHIN VSH Vp		0.834	10.8	7.7	13.3
A45-RF	RHOB PHIN VSH Vp		0.895	7.5	6.3	9.8
A45-XGBoost	RHOB PHIN VSH Vp		0.914	6.8	5.7	8.9

Appendix B. TD models for well-log combinations. R^2 , MAPE, SD, and RMSPE are calculated on the synthetic test dataset.

	Model	Well-logs	Coefficients	R^2	MAPE (%)	SD (%)	RMSPE (%)
Evaporites	B1-Linear	RHOB	$a_0 = -1.837, a_1 = 1.350$	0.266	25.7	19.3	32.1
	B1-AdaBoost	RHOB		0.270	25.3	19.0	31.6
	B1-RF	RHOB		0.289	24.8	18.9	31.2
	B1-XGBoost	RHOB		0.248	23.0	16.9	28.5
	B2-Linear	PHIN	$a_0 = 2.207, a_1 = -3.841$	0.833	11.6	9.6	15.1
	B2-AdaBoost	PHIN		0.873	10.0	7.9	12.8
	B2-RF	PHIN		0.892	8.7	7.0	11.1
	B2-XGBoost	PHIN		0.879	9.1	7.0	11.5
	B3-Linear	VSH	$a_0 = 1.477, a_1 = -0.821$	0.086	31.8	25.7	40.8
	B3-AdaBoost	VSH		0.087	32.0	25.7	40.9
	B3-RF	VSH		0.087	31.8	25.7	40.8
	B3-XGBoost	VSH		0.069	29.5	22.7	37.2
	B4-Linear	Vp	$a_0 = -0.472, a_1 = 0.426$	0.447	21.1	17.9	27.6
	B4-AdaBoost	Vp		0.461	20.9	17.9	27.5
	B4-RF	Vp		0.474	20.4	17.7	27.0
	B4-XGBoost	Vp		0.467	20.4	17.5	26.9
	B5-Linear	RHOB PHIN	$a_0 = 3.703, a_1 = -0.582, a_2 = -4.497$	0.859	11.2	9.9	14.9
	B5-AdaBoost	RHOB PHIN		0.892	9.9	8.1	12.8
	B5-RF	RHOB PHIN		0.914	8.1	6.3	10.3
	B5-XGBoost	RHOB PHIN		0.913	8.1	6.3	10.2
	B6-Linear	RHOB VSH	$a_0 = -1.543, a_1 = 1.311, a_2 = -0.742$	0.338	24.3	19.6	31.2
	B6-AdaBoost	RHOB VSH		0.305	24.7	19.6	31.6
	B6-RF	RHOB VSH		0.350	23.6	19.3	30.5
	B6-XGBoost	RHOB VSH		0.316	22.8	17.7	28.9
	B7-Linear	RHOB Vp	$a_0 = 0.551, a_1 = -0.718, a_2 = 0.579$	0.466	20.8	18.9	28.1
	B7-AdaBoost	RHOB Vp		0.471	20.9	18.0	27.5
	B7-RF	RHOB Vp		0.525	19.6	18.9	27.3
	B7-XGBoost	RHOB Vp		0.538	19.5	18.6	27.0
	B8-Linear	PHIN VSH	$a_0 = 2.273, a_1 = -3.743, a_2 = -0.325$	0.847	11.3	9.2	14.6
	B8-AdaBoost	PHIN VSH		0.876	10.1	8.2	13.0
	B8-RF	PHIN VSH		0.909	8.1	6.2	10.2
	B8-XGBoost	PHIN VSH		0.908	8.1	6.2	10.3
	B9-Linear	PHIN Vp	$a_0 = 2.667, a_1 = -4.301, a_2 = -0.089$	0.841	11.8	10.2	15.6
	B9-AdaBoost	PHIN Vp		0.887	10.0	8.3	13.0
	B9-RF	PHIN Vp		0.913	8.1	6.4	10.3
	B9-XGBoost	PHIN Vp		0.911	8.1	6.3	10.3

Carbonates	B10-Linear	VSH Vp	$a_0 = -0.236, a_1 = -0.641, a_2 = 0.411$	0.500	19.8	18.8	27.3
	B10-AdaBoost	VSH Vp		0.505	20.6	19.2	28.2
	B10-RF	VSH Vp		0.548	18.5	18.4	26.1
	B10-XGBoost	VSH Vp		0.543	18.6	18.4	26.1
	B11-Linear	RHOB PHIN VSH	$a_0 = 3.652, a_1 = -0.541., a_2 = -4.368, a_3 = -0.275$	0.869	10.9	9.4	14.4
	B11-AdaBoost	RHOB PHIN VSH		0.894	10.0	8.3	13.0
	B11-RF	RHOB PHIN VSH		0.831	7.3	5.7	9.3
	B11-XGBoost	RHOB PHIN VSH		0.835	7.2	5.5	9.0
	B12-Linear	RHOB PHIN Vp	$a_0 = 3.795, a_1 = -0.776, a_2 = -4.319, a_3 = 0.076$	0.862	10.9	9.5	14.5
	B12-AdaBoost	RHOB PHIN Vp		0.894	9.8	8.1	12.8
	B12-RF	RHOB PHIN Vp		0.916	8.0	6.2	10.2
	B12-XGBoost	RHOB PHIN Vp		0.919	7.9	6.1	10.0
	B13-Linear	RHOB VSH Vp	$a_0 = 0.673, a_1 = -0.643, a_2 = -0.620, a_3 = 0.549$	0.516	19.6	19.7	27.2
	B13-AdaBoost	RHOB VSH Vp		0.498	20.3	18.6	27.5
	B13-RF	RHOB VSH Vp		0.589	17.7	19.7	26.5
	B13-XGBoost	RHOB VSH Vp		0.618	16.8	19.9	26.1
	B14-Linear	PHIN VSH Vp	$a_0 = 2.680, a_1 = -4.152, a_2 = -0.304, a_3 = -0.077$	0.853	11.4	9.7	15.0
	B14-AdaBoost	PHIN VSH Vp		0.888	10.1	8.4	13.1
	B14-RF	PHIN VSH Vp		0.928	7.4	5.8	9.4
	B14-XGBoost	PHIN VSH Vp		0.938	6.8	5.5	8.7
	B15-Linear	RHOB PHIN VSH Vp	$a_0 = 3.747, a_1 = -0.740, a_2 = -4.182, a_3 = -0.277, a_4 = 0.078$	0.872	10.6	9.0	14.0
	B15-AdaBoost	RHOB PHIN VSH Vp		0.895	9.9	8.3	12.9
	B15-RF	RHOB PHIN VSH Vp		0.934	7.2	5.7	9.1
	B15-XGBoost	RHOB PHIN VSH Vp		0.950	6.1	5.1	8.0
	B16-Linear	RHOB	$a_0 = -2.477, a_1 = 1.493$	0.888	11.5	9.2	14.7
	B16-AdaBoost	RHOB		0.914	6.7	7.0	11.5
	B16-RF	RHOB		0.918	7.9	6.7	10.4
	B16-XGBoost	RHOB		0.918	7.9	6.7	10.3
	B17-Linear	PHIN	$a_0 = 1.703, a_1 = -2.445$	0.901	11.1	8.5	14.0
	B17-AdaBoost	PHIN		0.935	8.3	6.1	10.3
	B17-RF	PHIN		0.940	7.0	5.3	8.8
	B17-XGBoost	PHIN		0.940	7.0	5.2	8.7
	B18-Linear	VSH	$a_0 = 1.125, a_1 = -0.666$	0.039	41.8	31.3	52.3
	B18-AdaBoost	VSH		0.039	41.9	31.5	52.4
	B18-RF	VSH		0.039	41.8	13.3	52.2
	B18-XGBoost	VSH		0.021	39.0	27.3	47.6
	B19-Linear	Vp	$a_0 = -0.484, a_1 = 0.370$	0.944	6.7	4.9	8.3
	B19-AdaBoost	Vp		0.943	7.6	5.5	9.4
	B19-RF	Vp		0.951	5.9	4.5	7.5
	B19-XGBoost	Vp		0.951	5.9	4.5	7.4

B20-Linear	RHOB PHIN	$a_0 = 0.127, a_1 = 0.567, a_2 = -1.550$	0.909	10.8	8.4	13.6
B20-AdaBoost	RHOB PHIN		0.943	7.9	5.9	9.9
B20-RF	RHOB PHIN		0.951	6.2	4.9	7.9
B20-XGBoost	RHOB PHIN		0.953	6.0	4.8	7.7
B21-Linear	RHOB VSH	$a_0 = -2.346, a_1 = 1.477, a_2 = -0.319$	0.897	11.5	9.3	14.8
B21-AdaBoost	RHOB VSH		0.926	9.0	6.8	11.3
B21-RF	RHOB VSH		0.935	7.5	5.8	9.5
B21-XGBoost	RHOB VSH		0.936	7.3	5.7	9.3
B22-Linear	RHOB Vp	$a_0 = -0.814, a_1 = 0.236, a_2 = 0.316$	0.945	6.5	5.0	8.1
B22-AdaBoost	RHOB Vp		0.943	7.6	5.5	9.4
B22-RF	RHOB Vp		0.953	5.7	4.5	7.3
B22-XGBoost	RHOB Vp		0.955	5.7	4.4	7.2
B23-Linear	PHIN VSH	$a_0 = 1.768, a_1 = -2.420, a_2 = -0.248$	0.907	11.1	8.8	14.2
B23-AdaBoost	PHIN VSH		0.939	8.3	6.1	10.2
B23-RF	PHIN VSH		0.950	6.2	4.9	7.9
B23-XGBoost	PHIN VSH		0.951	6.1	4.9	7.9
B24-Linear	PHIN Vp	$a_0 = -0.039, a_1 = -0.517, a_2 = 0.297$	0.946	6.5	4.9	8.1
B24-AdaBoost	PHIN Vp		0.946	7.5	5.4	9.2
B24-RF	PHIN Vp		0.955	5.7	4.5	7.2
B24-XGBoost	PHIN Vp		0.956	5.6	4.5	7.2
B25-Linear	VSH Vp	$a_0 = -0.348, a_1 = -0.413, a_2 = 0.367$	0.958	5.9	4.6	7.5
B25-AdaBoost	VSH Vp		0.949	7.9	6.0	9.9
B25-RF	VSH Vp		0.969	4.7	3.8	6.0
B25-XGBoost	VSH Vp		0.969	4.7	3.8	6.0
B26-Linear	RHOB PHIN VSH	$a_0 = 0.085, a_1 = 0.607, a_2 = -1.458, a_3 = -0.271$	0.915	10.6	8.7	13.7
B26-AdaBoost	RHOB PHIN VSH		0.947	7.9	5.9	9.9
B26-RF	RHOB PHIN VSH		0.961	5.5	4.3	7.0
B26-XGBoost	RHOB PHIN VSH		0.972	4.6	3.6	5.8
B27-Linear	RHOB PHIN Vp	$a_0 = -0.209, a_1 = 0.072, a_2 = -0.437, a_3 = 0.291$	0.947	6.5	4.9	8.1
B27-AdaBoost	RHOB PHIN Vp		0.947	7.5	5.4	9.2
B27-RF	RHOB PHIN Vp		0.959	5.4	4.4	6.9
B27-XGBoost	RHOB PHIN Vp		0.964	5.0	4.1	6.5
B28-Linear	RHOB VSH Vp	$a_0 = -0.567, a_1 = 0.154, a_2 = -0.401, a_3 = 0.331$	0.959	6.0	4.5	7.5
B28-AdaBoost	RHOB VSH Vp		0.951	7.8	5.9	9.8
B28-RF	RHOB VSH Vp		0.972	4.5	3.7	5.8
B28-XGBoost	RHOB VSH Vp		0.974	4.2	3.5	5.5
B29-Linear	PHIN VSH Vp	$a_0 = -0.108, a_1 = -0.286, a_2 = -0.392, a_3 = 0.326$	0.959	6.0	4.6	7.5
B29-AdaBoost	PHIN VSH Vp		0.950	7.8	6.0	9.9
B29-RF	PHIN VSH Vp		0.969	4.7	3.8	6.0
B29-XGBoost	PHIN VSH Vp		0.970	4.5	3.7	5.9

B30-Linear	RHOB PHIN VSH Vp	$a_0 = -0.303, a_1 = 0.083, a_2 = -0.195, a_3 = -0.392, a_4 = 0.320$	0.959	6.0	4.5	7.5
B30-AdaBoost	RHOB PHIN VSH Vp		0.952	7.7	6.0	9.8
B30-RF	RHOB PHIN VSH Vp		0.972	4.5	3.6	5.8
B30-XGBoost	RHOB PHIN VSH Vp		0.980	3.8	3.1	4.9
B31-Linear	RHOB	$a_0 = -1.993, a_1 = 1.254$	0.666	19.6	15.6	25.1
B31-AdaBoost	RHOB		0.675	18.3	14.9	23.6
B31-RF	RHOB		0.688	17.5	14.6	22.8
B31-XGBoost	RHOB		0.688	17.1	13.4	21.7
B32-Linear	PHIN	$a_0 = 1.529, a_1 = -2.224$	0.790	16.1	11.1	19.6
B32-AdaBoost	PHIN		0.866	12.3	8.9	15.2
B32-RF	PHIN		0.878	10.7	8.1	13.4
B32-XGBoost	PHIN		0.878	10.7	7.9	13.3
B33-Linear	VSH	$a_0 = 1.214, a_1 = -0.717$	0.147	40.9	31.2	51.5
B33-AdaBoost	VSH		0.152	40.9	31.0	51.3
B33-RF	VSH		0.153	40.8	30.9	51.1
B33-XGBoost	VSH		0.153	40.5	30.6	50.8
B34-Linear	Vp	$a_0 = -0.554, a_1 = 0.383$	0.762	14.6	10.8	18.2
B34-AdaBoost	Vp		0.754	15.1	11.3	18.9
B34-RF	Vp		0.771	14.2	10.8	17.8
B34-XGBoost	Vp		0.771	13.9	10.3	17.3
B35-Linear	RHOB PHIN	$a_0 = 1.716, a_1 = -0.069, a_2 = -2.327$	0.790	16.1	11.2	19.6
B35-AdaBoost	RHOB PHIN		0.867	12.3	8.9	15.2
B35-RF	RHOB PHIN		0.887	10.0	8.1	12.9
B35-XGBoost	RHOB PHIN		0.791	9.7	8.0	12.5
B36-Linear	RHOB VSH	$a_0 = -1.576, a_1 = 1.210, a_2 = -0.577$	0.757	18.7	14.8	23.8
B36-AdaBoost	RHOB VSH		0.816	16.0	12.6	20.3
B36-RF	RHOB VSH		0.843	13.2	10.6	17.0
B36-XGBoost	RHOB VSH		0.844	13.2	10.6	16.9
B37-Linear	RHOB Vp	$a_0 = -0.606, a_1 = 0.040, a_2 = 0.373$	0.762	14.6	10.9	18.2
B37-AdaBoost	RHOB Vp		0.761	15.2	11.1	18.8
BA37-RF	RHOB Vp		0.790	13.5	10.9	17.3
B37-XGBoost	RHOB Vp		0.793	13.4	10.8	17.2
B38-Linear	PHIN VSH	$a_0 = 1.752, a_1 = -2.128, a_2 = -0.458$	0.846	14.6	13.3	19.7
B38-AdaBoost	PHIN VSH		0.896	11.6	8.5	14.4
B38-RF	PHIN VSH		0.924	8.0	6.8	10.6
B38-XGBoost	PHIN VSH		0.925	8.0	6.8	10.5
B39-Linear	PHIN Vp	$a_0 = 0.968, a_1 = -1.600, a_2 = 0.105$	0.794	15.2	10.3	18.4
B39-AdaBoost	PHIN Vp		0.866	12.3	8.9	15.2
B39-RF	PHIN Vp		0.891	9.6	8.1	12.6
B39-XGBoost	PHIN Vp		0.889	9.6	8.1	12.6

B40-Linear	VSH Vp	$a_0 = -0.193, a_1 = -0.578, a_2 = 0.372$	0.853	13.0	10.4	16.7
B40-AdaBoost	VSH Vp		0.865	12.9	9.5	16.0
B40-RF	VSH Vp		0.924	8.0	6.9	10.6
B40-XGBoost	VSH Vp		0.925	7.9	6.8	10.4
B41-Linear	RHOB PHIN VSH	$a_0 = 1.567, a_1 = 0.069, a_2 = -2.024, a_3 = -0.463$	0.846	14.6	13.3	19.7
B41-AdaBoost	RHOB PHIN VSH		0.900	11.4	8.4	14.2
B41-RF	RHOB PHIN VSH		0.927	7.9	6.7	10.3
B41-XGBoost	RHOB PHIN VSH		0.930	7.6	6.5	10.0
B42-Linear	RHOB PHIN Vp	$a_0 = 1.350, a_1 = -0.201, a_2 = -1.779, a_3 = 0.136$	0.796	14.9	10.2	18.1
B42-AdaBoost	RHOB PHIN Vp		0.866	12.1	8.7	14.9
B42-RF	RHOB PHIN Vp		0.907	8.9	7.6	11.7
B42-XGBoost	RHOB PHIN Vp		0.915	8.4	7.2	11.0
B43-Linear	RHOB VSH Vp	$a_0 = -0.187, a_1 = -0.004, a_2 = -0.578, a_3 = 0.373$	0.853	13.0	10.4	16.6
B43-AdaBoost	RHOB VSH Vp		0.866	12.8	9.4	15.9
B43-RF	RHOB VSH Vp		0.924	7.9	6.8	10.5
B43-XGBoost	RHOB VSH Vp		0.928	7.8	6.7	10.3
B44-Linear	PHIN VSH Vp	$a_0 = 0.649, a_1 = -0.949, a_2 = -0.522, a_3 = 0.213$	0.863	13.0	11.2	17.1
B44-AdaBoost	PHIN VSH Vp		0.900	11.4	8.4	14.1
B44-RF	PHIN VSH Vp		0.934	7.6	6.4	10.0
B44-XGBoost	PHIN VSH Vp		0.937	7.4	6.3	9.7
B45-Linear	RHOB PHIN VSH Vp	$a_0 = 0.920, a_1 = -0.142, a_2 = -1.047, a_3 = -0.518, a_4 = 0.234$	0.863	12.9	10.9	16.9
B45-AdaBoost	RHOB PHIN VSH Vp		0.900	11.4	8.4	14.1
B45-RF	RHOB PHIN VSH Vp		0.936	7.6	6.4	9.9
B45-XGBoost	RHOB PHIN VSH Vp		0.946	6.9	5.9	9.1

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Appendix C. SHC models for well-log combinations. R^2 , MAPE, SD, and RMSPE are calculated on the synthetic test dataset.

	Model	Well-logs	Coefficients	R^2	MAPE (%)	SD (%)	RMSPE (%)
Evaporites	C1-Linear	RHOB	$a_0 = 4599.8, a_1 = -1432.8$	0.671	11.9	10.2	15.6
	C1-AdaBoost	RHOB		0.676	11.7	9.8	15.3
	C1-RF	RHOB		0.702	10.9	9.8	14.7
	C1-XGBoost	RHOB		0.682	11.5	9.9	15.1
	C2-Linear	PHIN	$a_0 = 705.3, a_1 = 2478.9$	0.767	9.8	8.5	13.0
	C2-AdaBoost	PHIN		0.777	9.6	8.2	12.6
	C2-RF	PHIN		0.810	8.5	8.1	11.8
	C2-XGBoost	PHIN		0.783	9.4	8.3	12.5
	C3-Linear	VSH	$a_0 = 1325.8, a_1 = -10.2$	0.001	22.5	16.8	28.1
	C3-AdaBoost	VSH		0.001	22.5	16.8	28.1
	C3-RF	VSH		0.001	22.5	16.8	28.1
	C3-XGBoost	VSH		0.001	21.5	16.8	28.1
	C4-Linear	Vp	$a_0 = 2954.4, a_1 = -403.3$	0.894	6.9	6.4	9.4
	C4-AdaBoost	Vp		0.937	5.3	4.4	6.9
	C4-RF	Vp		0.942	4.9	4.2	6.5
	C4-XGBoost	Vp		0.941	5.1	4.2	6.5
	C5-Linear	RHOB PHIN	$a_0 = 2535.4, a_1 = -712.7, a_2 = 1675.7$	0.854	7.9	6.7	10.3
	C5-AdaBoost	RHOB PHIN		0.849	8.2	6.3	10.3
	C5-RF	RHOB PHIN		0.876	7.1	6.1	9.3
	C5-XGBoost	RHOB PHIN		0.875	7.1	6.0	9.3
	C6-Linear	RHOB VSH	$a_0 = 4638.3, a_1 = -1437.9, a_2 = -97.1$	0.674	11.8	10.1	15.5
	C6-AdaBoost	RHOB VSH		0.676	11.7	9.9	15.3
	C6-RF	RHOB VSH		0.701	11.0	9.8	14.7
	C6-XGBoost	RHOB VSH		0.687	11.1	9.8	14.8
	C7-Linear	RHOB Vp	$a_0 = 2905.1, a_1 = 34.6, a_2 = -410.7$	0.894	6.9	9.8	9.4
	C7-AdaBoost	RHOB Vp		0.937	5.3	4.3	6.8
	C7-RF	RHOB Vp		0.947	4.5	4.2	6.2
	C7-XGBoost	RHOB Vp		0.947	4.5	4.2	6.1
	C8-Linear	PHIN VSH	$a_0 = 776.0, a_1 = 2585.7, a_2 = -353.0$	0.801	8.8	8.0	11.9
	C8-AdaBoost	PHIN VSH		0.789	9.3	7.9	12.2
	C8-RF	PHIN VSH		0.844	7.6	7.6	10.8
	C8-XGBoost	PHIN VSH		0.843	7.8	7.7	11.8
	C9-Linear	PHIN Vp	$a_0 = 2242.6, a_1 = 972.4, a_2 = -287.3$	0.938	5.3	4.9	7.2
	C9-AdaBoost	PHIN Vp		0.951	4.7	3.8	6.1
	C9-RF	PHIN Vp		0.965	3.8	3.4	5.1
	C9-XGBoost	PHIN Vp		0.966	3.7	3.3	5.0

Carbonates	C10-Linear	VSH Vp	$a_0 = 3023.7, a_1 = -188.4, a_2 = -407.6$	0.904	6.5	6.1	8.9
	C10-AdaBoost	VSH Vp		0.938	5.2	4.3	6.8
	C10-RF	VSH Vp		0.950	4.6	4.0	6.1
	C10-XGBoost	VSH Vp		0.950	4.6	4.0	6.1
	C11-Linear	RHOB PHIN VSH	$a_0 = 2481.6, a_1 = -669.0, a_2 = 1813.0, a_3 = -291.0$	0.877	7.0	6.2	9.4
	C11-AdaBoost	RHOB PHIN VSH		0.854	8.0	6.1	10.0
	C11-RF	RHOB PHIN VSH		0.902	6.1	5.6	8.3
	C11-XGBoost	RHOB PHIN VSH		0.915	5.6	5.4	7.8
	C12-Linear	RHOB PHIN Vp	$a_0 = 2173.8, a_1 = 47.7, a_2 = 973.4, a_3 = -297.3$	0.938	5.2	4.9	7.2
	C12-AdaBoost	RHOB PHIN Vp		0.951	4.7	3.8	6.1
	C12-RF	RHOB PHIN Vp		0.968	3.4	3.4	4.8
	C12-XGBoost	RHOB PHIN Vp		0.970	3.3	3.3	4.7
	C13-Linear	RHOB VSH Vp	$a_0 = 2942.6, a_1 = 57.4, a_2 = -190.3, a_3 = -419.9$	0.904	6.5	6.1	8.9
	C13-AdaBoost	RHOB VSH Vp		0.939	5.2	4.3	6.7
	C13-RF	RHOB VSH Vp		0.956	4.0	4.0	5.7
	C13-XGBoost	RHOB VSH Vp		0.958	3.8	4.0	5.5
	C14-Linear	PHIN VSH Vp	$a_0 = 2245.0, a_1 = 1109.0, a_2 = -278.4, a_3 = -277.3$	0.958	4.2	4.0	5.8
	C14-AdaBoost	PHIN VSH Vp		0.966	3.9	3.0	4.9
	C14-RF	PHIN VSH Vp		0.983	2.6	2.5	3.6
	C14-XGBoost	PHIN VSH Vp		0.989	1.9	2.2	2.9
	C15-Linear	RHOB PHIN VSH Vp	$a_0 = 2125.0, a_1 = 83.3, a_2 = 1112.3, a_3 = -281.5, a_4 = -294.7$	0.959	4.2	4.0	5.8
	C15-AdaBoost	RHOB PHIN VSH Vp		0.967	3.9	3.1	5.0
	C15-RF	RHOB PHIN VSH Vp		0.985	2.3	2.5	3.4
	C15-XGBoost	RHOB PHIN VSH Vp		0.993	1.4	1.8	2.3
	C16-Linear	RHOB	$a_0 = 5984.9, a_1 = -1910.2$	0.951	6.2	6.5	9.0
	C16-AdaBoost	RHOB		0.952	6.3	5.7	8.5
	C16-RF	RHOB		0.958	5.6	5.8	8.0
	C16-XGBoost	RHOB		0.956	5.8	5.8	8.2
	C17-Linear	PHIN	$a_0 = 630.8, a_1 = 3147.5$	0.978	4.2	4.1	5.9
	C17-AdaBoost	PHIN		0.977	4.6	3.6	5.8
	C17-RF	PHIN		0.982	3.5	3.7	5.1
	C17-XGBoost	PHIN		0.980	3.9	3.8	5.4
	C18-Linear	VSH	$a_0 = 1617.7, a_1 = 36.1$	0.001	31.0	25.8	40.4
	C18-AdaBoost	VSH		0.001	31.0	25.8	40.3
	C18-RF	VSH		0.001	31.0	25.8	40.4
	C18-XGBoost	VSH		0.001	31.0	25.8	40.3
	C19-Linear	Vp	$a_0 = 3362.1, a_1 = -455.8$	0.930	7.7	7.3	10.6
	C19-AdaBoost	Vp		0.992	2.0	2.3	3.1
	C19-RF	Vp		0.997	1.3	2.1	2.5
	C19-XGBoost	Vp		0.996	1.4	2.2	2.6

C20-Linear	RHOB PHIN	$a_0 = 2035.6, a_1 = -505.0, a_2 = 2349.4$	0.982	3.9	3.8	5.4
C20-AdaBoost	RHOB PHIN		0.979	4.4	3.4	5.6
C20-RF	RHOB PHIN		0.986	3.1	3.4	4.6
C20-XGBoost	RHOB PHIN		0.988	2.5	3.5	4.3
C21-Linear	RHOB VSH	$a_0 = 6158.1, a_1 = -1931.9, a_2 = -419.0$	0.962	5.6	5.7	8.0
C21-AdaBoost	RHOB VSH		0.954	6.2	5.4	8.2
C21-RF	RHOB VSH		0.964	5.2	5.1	7.3
C21-XGBoost	RHOB VSH		0.965	5.2	5.1	7.3
C22-Linear	RHOB Vp	$a_0 = 5080.0, a_1 = -1226.0, a_2 = -171.9$	0.962	5.5	5.8	8.0
C22-AdaBoost	RHOB Vp		0.992	2.1	2.4	3.2
C22-RF	RHOB Vp		0.997	1.3	2.0	2.4
C22-XGBoost	RHOB Vp		0.997	1.3	2.0	2.4
C23-Linear	PHIN VSH	$a_0 = 766.7, a_1 = 3199.5, a_2 = -516.8$	0.994	2.2	2.3	3.2
C23-AdaBoost	PHIN VSH		0.980	4.0	2.9	5.9
C23-RF	PHIN VSH		0.995	1.6	2.2	2.7
C23-XGBoost	PHIN VSH		0.995	1.6	2.2	2.7
C24-Linear	PHIN Vp	$a_0 = 1109.9, a_1 = 2617.4, a_2 = -81.6$	0.980	4.0	4.1	5.7
C24-AdaBoost	PHIN Vp		0.992	2.1	2.3	3.1
C24-RF	PHIN Vp		0.997	1.0	2.1	2.3
C24-XGBoost	PHIN Vp		0.997	1.0	2.2	2.4
C25-Linear	VSH Vp	$a_0 = 3454.1, a_1 = -279.9, a_2 = -457.2$	0.935	5.0	6.8	10.1
C25-AdaBoost	VSH Vp		0.992	2.0	2.3	3.1
C25-RF	VSH Vp		0.999	0.8	0.9	1.2
C25-XGBoost	VSH Vp		0.999	0.7	0.7	1.0
C26-Linear	RHOB PHIN VSH	$a_0 = 1957.7, a_1 = -429.7, a_2 = 2518.9, a_3 = -500.4$	0.996	1.7	1.7	2.4
C26-AdaBoost	RHOB PHIN VSH		0.986	3.4	2.7	4.4
C26-RF	RHOB PHIN VSH		0.998	0.9	1.6	1.9
C26-XGBoost	RHOB PHIN VSH		0.999	0.6	1.0	1.2
C27-Linear	RHOB PHIN Vp	$a_0 = 2094.2, a_1 = -418.9, a_2 = 2155.6, a_3 = -50.8$	0.983	3.8	3.8	5.4
C27-AdaBoost	RHOB PHIN Vp		0.992	2.1	2.4	3.2
C27-RF	RHOB PHIN Vp		0.998	0.8	2.0	2.1
C27-XGBoost	RHOB PHIN Vp		0.998	0.5	1.9	2.0
C28-Linear	RHOB VSH Vp	$a_0 = 5313.9, a_1 = -1303.9, a_2 = -379.8, a_3 = -157.3$	0.971	4.9	5.0	7.0
C28-AdaBoost	RHOB VSH Vp		0.996	1.6	2.3	2.8
C28-RF	RHOB VSH Vp		0.999	0.8	0.9	1.2
C28-XGBoost	RHOB VSH Vp		0.999	0.6	0.7	0.9
C29-Linear	PHIN VSH Vp	$a_0 = 1021.2, a_1 = 2910.3, a_2 = -497.4, a_3 = -44.2$	0.994	2.2	2.2	3.1
C29-AdaBoost	PHIN VSH Vp		0.992	2.1	2.4	3.2
C29-RF	PHIN VSH Vp		0.999	0.3	0.4	0.5
C29-XGBoost	PHIN VSH Vp		0.999	0.2	0.2	0.3

Clastic rocks	C30-Linear	RHOB PHIN VSH Vp	$a_0 = 1975.3, a_1 = -405.9, a_2 = 2461.5, a_3 = -494.9, a_4 = -14.6$	0.996	1.7	1.7	2.4
	C30-AdaBoost	RHOB PHIN VSH Vp		0.992	2.1	2.3	3.1
	C30-RF	RHOB PHIN VSH Vp		0.999	0.3	0.4	0.5
	C30-XGBoost	RHOB PHIN VSH Vp		0.999	0.1	0.1	0.2
	C31-Linear	RHOB	$a_0 = 5826.4, a_1 = -1900.5$	0.896	10.0	10.3	14.3
	C31-AdaBoost	RHOB		0.898	9.9	9.2	13.5
	C31-RF	RHOB		0.911	8.8	9.2	12.8
	C31-XGBoost	RHOB		0.905	9.3	9.3	13.2
	C32-Linear	PHIN	$a_0 = 538.9, a_1 = 3212.5$	0.969	5.5	5.6	7.9
	C32-AdaBoost	PHIN		0.968	5.8	5.0	7.6
	C32-RF	PHIN		0.975	4.6	4.9	6.7
	C32-XGBoost	PHIN		0.972	5.0	4.9	7.0
	C33-Linear	VSH	$a_0 = 1555.1, a_1 = 22.1$	0.001	34.0	29.3	44.9
	C33-AdaBoost	VSH		0.001	34.0	29.3	44.9
	C33-RF	VSH		0.001	34.0	29.3	44.9
	C33-XGBoost	VSH		0.001	34.0	29.3	44.9
	C34-Linear	Vp	$a_0 = 3549.8, a_1 = -554.0$	0.936	8.1	8.2	11.5
	C34-AdaBoost	Vp		0.985	3.9	3.7	5.4
	C34-RF	Vp		0.992	2.4	3.4	4.2
	C34-XGBoost	Vp		0.992	2.5	3.4	4.2
	C35-Linear	RHOB PHIN	$a_0 = 1916.8, a_1 = -506.3, a_2 = 2452.9$	0.978	4.7	4.7	6.6
Clastic rocks	C35-AdaBoost	RHOB PHIN		0.976	5.0	4.3	6.6
	C35-RF	RHOB PHIN		0.985	3.5	3.9	5.2
	C35-XGBoost	RHOB PHIN		0.986	3.3	3.9	5.1
	C36-Linear	RHOB VSH	$a_0 = 5970.9, a_1 = -1915.6, a_2 = -199.9$	0.902	9.7	10.0	13.9
	C36-AdaBoost	RHOB VSH		0.901	9.8	8.8	13.2
	C36-RF	RHOB VSH		0.914	8.6	8.8	12.3
	C36-XGBoost	RHOB VSH		0.914	8.6	8.8	12.3
	C37-Linear	RHOB Vp	$a_0 = 4448.5, a_1 = -694.7, a_2 = -370.1$	0.952	6.8	7.3	10.0
	C37-AdaBoost	RHOB Vp		0.984	4.1	3.7	5.5
	C37-RF	RHOB Vp		0.992	2.3	3.4	4.1
	C37-XGBoost	RHOB Vp		0.992	2.3	3.4	4.1
	C38-Linear	PHIN VSH	$a_0 = 723.7, a_1 = 3292.2, a_2 = -380.0$	0.991	2.9	3.0	4.2
	C38-AdaBoost	PHIN VSH		0.978	4.5	3.9	6.0
	C38-RF	PHIN VSH		0.995	1.9	2.6	3.2
	C38-XGBoost	PHIN VSH		0.995	1.9	2.6	3.2
	C39-Linear	PHIN Vp	$a_0 = 1386.5, a_1 = 2341.4, a_2 = -158.9$	0.975	5.0	5.3	7.3
	C39-AdaBoost	PHIN Vp		0.984	4.1	3.5	5.4
	C39-RF	PHIN Vp		0.993	2.2	3.3	3.9
	C39-XGBoost	PHIN Vp		0.993	2.2	3.3	3.9

C40-Linear	VSH Vp	$a_0 = 3667.1, a_1 = -187.7, a_2 = -557.7$	0.941	7.7	7.5	10.8
C40-AdaBoost	VSH Vp		0.984	4.1	3.6	5.4
C40-RF	VSH Vp		0.997	1.2	2.3	2.6
C40-XGBoost	VSH Vp		0.997	1.1	2.1	2.5
C41-Linear	RHOB PHIN VSH	$a_0 = 1803.2, a_1 = -401.7, a_2 = 2683.9, a_3 = -352.2$	0.997	1.7	1.8	2.4
C41-AdaBoost	RHOB PHIN VSH		0.983	3.9	3.1	5.0
C41-RF	RHOB PHIN VSH		0.998	1.0	1.6	1.9
C41-XGBoost	RHOB PHIN VSH		0.999	0.6	1.1	1.2
C42-Linear	RHOB PHIN Vp	$a_0 = 2172.1, a_1 = -414.2, a_2 = 2071.0, a_3 = -94.9$	0.980	4.5	4.6	6.5
C42-AdaBoost	RHOB PHIN Vp		0.984	4.1	3.6	5.4
C42-RF	RHOB PHIN Vp		0.994	1.9	3.1	3.7
C42-XGBoost	RHOB PHIN Vp		0.995	1.6	3.0	3.4
C43-Linear	RHOB VSH Vp	$a_0 = 4593.1, a_1 = -710.1, a_2 = -199.4, a_3 = -370.0$	0.958	6.4	6.5	9.1
C43-AdaBoost	RHOB VSH Vp		0.984	4.1	3.5	5.4
C43-RF	RHOB VSH Vp		0.997	1.2	2.3	2.6
C43-XGBoost	RHOB VSH Vp		0.997	1.2	2.2	2.5
C44-Linear	PHIN VSH Vp	$a_0 = 1169.5, a_1 = 2815.7, a_2 = -354.1, a_3 = -86.0$	0.993	2.6	2.6	3.8
C44-AdaBoost	PHIN VSH Vp		0.986	3.5	3.5	4.2
C44-RF	PHIN VSH Vp		0.999	0.8	0.8	1.3
C44-XGBoost	PHIN VSH Vp		0.999	0.7	0.7	1.1
C45-Linear	RHOB PHIN VSH Vp	$a_0 = 1885.9, a_1 = -374.7, a_2 = 2559.0, a_3 = -345.1, a_4 = -29.9$	0.997	1.6	1.6	2.4
C45-AdaBoost	RHOB PHIN VSH Vp		0.987	3.5	3.5	4.2
C45-RF	RHOB PHIN VSH Vp		0.999	0.8	0.8	1.3
C45-XGBoost	RHOB PHIN VSH Vp		0.999	0.5	0.5	0.7

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